

Abstract

This study investigates the influence of private equity investment on innovation within the US pharmaceutical sector, utilizing firm-level data from 1985 to 2023. The study examines the impact of private equity participation on innovation production, innovation value, research and development intensity, and financial health. Utilizing t-tests, ordinary least squares regressions, and random effects panel regressions to understand the impact of private equity investment. The findings indicate that private equity investment boosts innovation output, increases innovation value, and increases R&D intensity. Finding private equity has no dampening effect on R&D intensity caused by financial leverage (debt) and no impact on pharmaceutical financial health.

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1. Introduction

This study's findings expand upon existing literature regarding the influence of private equity investment on innovation, expanding the research into the US pharmaceutical sector. Over the past four decades, private equity (PE) has emerged as a significant contributor in the pharmaceutical industry; therefore, this dissertation examines the influence of private equity backing on innovation output and value. It expands to investigate whether PE support influences research and development expenditures and firm financial health. The study was inspired by the absence of sector-specific analysis into private equity's impact on pharmaceuticals.

Private equity's emergence in the US pharmaceutical sector can be linked to sector-specific characteristics. The pharmaceutical industry is recession-resistant, has vast opportunities to produce high returns, and is highly fragmented (Cohen, 2019). Existing literature highlights private equity's frequent focus on subsectors, including contract research organizations, contract development and manufacturing organizations, generic medication manufacturers, and specialized pharmaceutical firms (Dabbs, 2019).

Existing literature has found that PE firms boost the value and output of innovation in pharmaceuticals (Amess et al., 2016). PE has boosted innovation by introducing value development methodologies and operational modifications. The current literature fails to encompass PE's influence on innovation within pharmaceuticals. The identified gaps provide the foundation for creating the primary research questions and hypotheses that direct this dissertation.

The analysis methodology used by Amess et al. (2016) serves as the groundwork for assessing the impact of PE investment on innovation. The dataset was built utilizing multiple databases. Existing literature guided the selection of variables for the regression models. T-tests, ordinary least squares regressions, and random effects panel regressions were used to assess the influence of PE-backing. The primary variables identified were innovation output (patent counts), innovation value (forward patent citations), R&D intensity, the interplay between leverage and R&D, and financial health.

The empirical results of the regressions aligned primarily with the existing literature. The regressions and t-tests indicated that PE investment enhances innovation output, value, and R&D

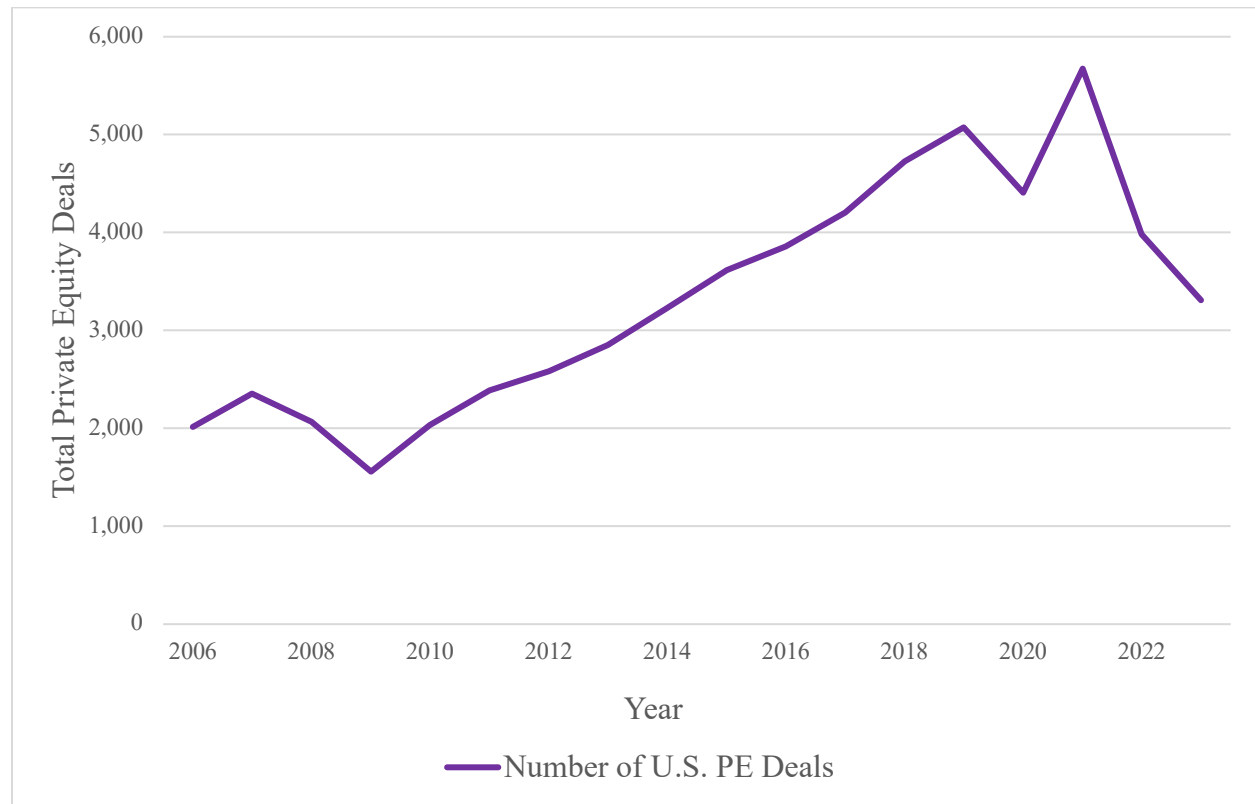
intensity. PE investment has no statistically significant impact on financial health and does not impact the relationship between leverage and R&D expenditure. Robustness tests were employed to validate the results. The obtained results were hindered by methodological constraints and model misspecification. The results of the empirical analysis set the stage for additional research into the firm-level changes private equity brings to pharmaceuticals.

2. Background

2.1 Growth of Private Equity Investment in Pharmaceutical Companies

Private equity investment in all US sectors has risen for the past four decades; deal value increased by 36% between 2023 and 2024 (Witte, 2024). Deal volume has skyrocketed from the early 2000s, with over 3000 deals occurring in 2023 (Graph 1) (Statista, 2024). Private equity firms have recognized the significant growth opportunities present in the healthcare industry. US healthcare expenditure totaled \$4 trillion in 2024, representing 17% of the US GDP (Daggett, 2024). The pharmaceutical industry in the US employs over 338 thousand people, generating over \$310 billion in revenue in 2024 (Brocker, 2024b) (Brocker, 2024a). The pharmaceutical industry in the US is poised to grow 5-10% a year between 2023 and 2028 (Singhal et al., 2025). The key drivers of the growth are an aging population and rising chronic disease in the US (Neveling, 2023).

Graph 1: US PE Deal Volume (Statista, 2024)



Before the 1990s, PE activity in the pharmaceutical industry was minimal (Walton, 2000). The invention of penicillin during the Second World War sparked a period of rapid innovation in the pharmaceutical industry (Younkin, 2008). Pharmaceutical firms invested heavily in research and development (R&D), introducing many new chemically unique drugs between the 1940s and early 1980s (Busfield, 2020). The 1980s brought a slowdown in the industry, with many firms enjoying high profits on patented drugs, preferring to dedicate R&D to developing chemically similar drugs, known as “me-too drugs” (Angell and Angell, 2005) (LIGHT and Lexchin, 2012).

The slowdown in innovation was met with increased M&A and PE activity in the 1990s (Busfield, 2020). PE recognized the growth opportunities within the pharmaceutical industry and rapidly increased investment. In the past decade, PE investment in healthcare alone has amounted to \$750 billion within the US (Scheffler Richard M and Alexander, 2021). The fundamentals of the

pharmaceutical industry are ripe for private equity managers to take advantage of. Between 1997 and 2025, 4,945 deals were made by PE firms with pharmaceutical companies (M&A, 2025).

2.2 Drivers of Private Equity Investment in the Pharmaceutical Sector

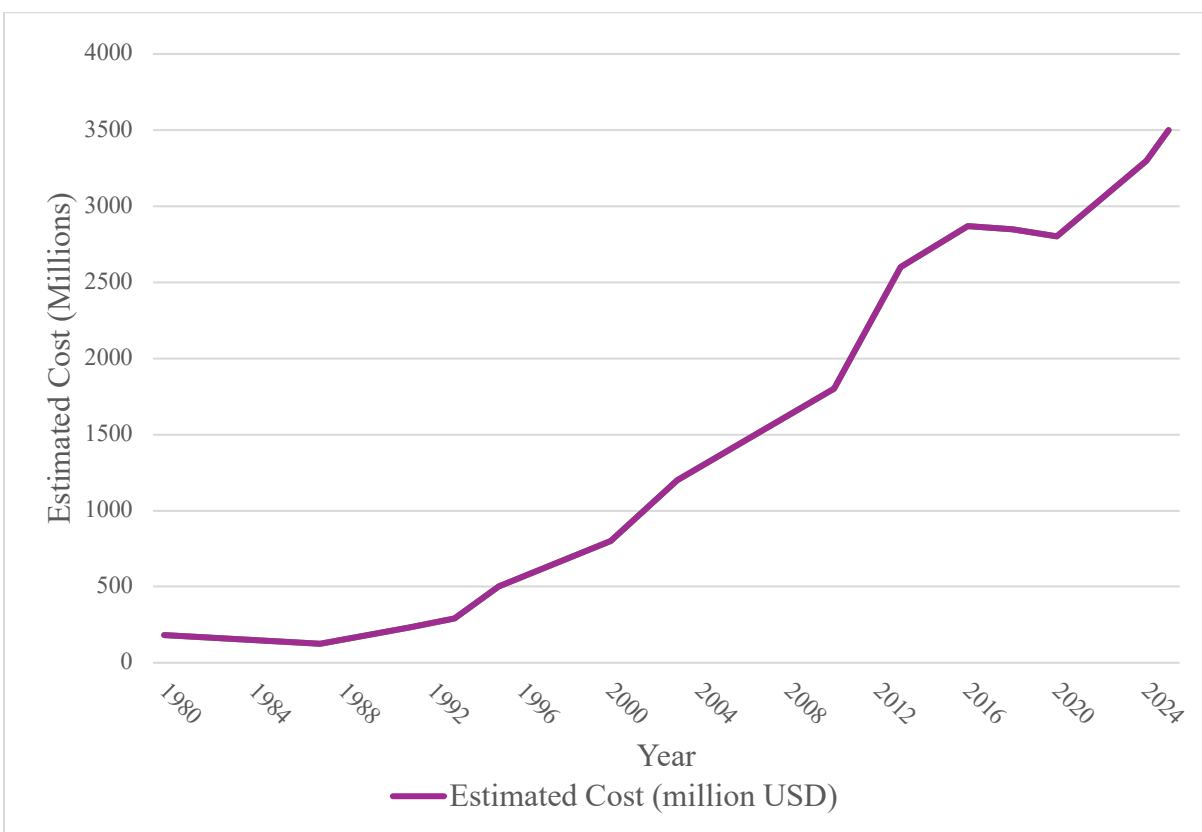
Pharmaceutical companies have relatively inelastic demand, as their products are often necessities for consumers (Cohen, 2019). During the 2008-2009 financial crisis, large pharmaceutical firms like Pfizer and Bristol-Myers Squibb experienced sales growth of 7% and 8%, respectively (Cohen, 2019). Pharmaceutical companies are less exposed to macroeconomic factors such as GDP, inflation, employment, and fiscal policy changes (O'Brien, 2025). During recessions, the pharmaceutical sector's earnings decrease by half compared to the rest of the market (O'Brien, 2025). PE firms aim to acquire companies with good growth potential and profitability using measures such as the gross internal rate of return to value firms, which favor firms with lower risk (Gompers et al., 2016) (Goold, 2007). This recession-resistant nature of pharmaceutical firms makes them ideal for PE investment.

Private equity invests in pharmaceuticals due to their consistently high profit margins. The pharmaceutical sector has exceptionally high returns, with a net profit margin of 15.6% on branded products and 6% on generic products in 2024 (Brocker, 2024a, Brocker, 2024b). Competition in the generic drug sector is greater due to the lower barriers to entry; hence, profit margins are lower (Brocker, 2024b). For this reason, PE tends to prefer investing in branded pharmaceuticals over generics (Rand, 2021). Pharmaceuticals also have low revenue volatility, providing consistent returns (Brocker, 2024a).

The pharmaceutical industry is fragmented, predominantly consisting of small and medium enterprises (SMEs) (Daggett, 2024). Private equity firms seek to acquire stakes in firms with ample growth opportunities and that are within a fragmented market (Babus and Parlatore, 2022). Innovation within the industry is also fragmented, with small and medium firms engaging in the most innovation; hence, they possess the best growth opportunities in the sector (Busfield, 2020). Drug development costs are consistently increasing, with the drug development and approval process costing over \$2.87 billion in 2016 (DiMasi et al., 2016) (Graph 2). This high associated cost of developing drugs results in most firms requiring external support and seeking investment

from capital markets. PE firms step in to fill this void, targeting SMEs in the pharmaceutical sector specifically due to the fragmentation and greater growth opportunities (Andersson et al., 2010).

Graph 2: Drug Price Development Cost over Time (data from Tufts, 2025)



2.3 Private Equity's Strategic Focus Areas

Drug development in the pharmaceutical industry has split, with pharmaceutical firms contracting drug development and research to external firms. Contract research organizations (CROs) and contract development and manufacturing organizations (CDMOs) are tasked with researching, trialing, and developing drugs on behalf of pharmaceutical companies (Scientific, 2025) (Pharmaceuticals, 2025). The CRO and CDMO sector is highly fragmented, serviced by predominantly smaller firms, providing private equity with an easier entry point into the industry (Floersh, 2024). CROs and CDMOs are the leading players in bringing new drugs for cell and gene therapies, immunotherapies, and mRNA-based vaccines to market (Floersh, 2024). CROs and

CDMOs are paid on a fixed contract basis per drug (Moreira et al., 2023). PE invests in firms with ample growth opportunities and substantial returns (Jose Luis Blanco and 2024). PE targets these CROs and CDMOs due to their fragmented nature, expansive growth opportunities, and consistent revenue from contracts.

The biopharmaceutical sector has continued to experience rapid growth, with most pharmaceutical innovation now originating from advances within this field. Biopharma accounted for 73% of late-stage drugs in 2018 and 64% of new drugs launched in 2018 (Aitken et al., 2022). Most developments are chemically unique products, often patented, giving the firm exclusivity for their production (Aitken et al., 2022). BioPharma's strong growth prospects and potential for high returns have made these companies attractive targets for PE investment (Andersson et al., 2010). PE has approached biopharma at two different levels of development: early-stage and late-stage (Dabbs, 2019). PE's entry point depends on the suspected growth characteristics of the firm. Early-stage entries are typically high risk; PE has usually preferred to take on minority stakes in these firms (Dabbs, 2019). Late-stage firms are often publicly traded; PE frequently acquires and delists these firms, seeking to introduce internal changes before final drug delivery (Dabbs, 2019).

3. Literature Review

3.1 Private Equity Impact on Patenting

Literature on innovation has found that public sector investment in private biomedicine significantly boosts patenting activity. The National Institute of Health (NIH) provided grants of ten million USD to different private sector biopharma firms, producing an average of 2.3–2.7 more patents (Azoulay et al., 2019). These grants focused on supporting research and development carried out by private companies. The public sector has been a source of funding to help encourage innovation while lowering risk and expenses for private companies. NIH-funded patents sometimes center on incremental innovations rather than groundbreaking ideas (Azoulay et al., 2019). Fernald et al. (2024) argue that pharmaceutical firms often support incremental improvements due to the lower developmental risk.

A range of studies have highlighted differing results regarding the impact of private equity backing on innovative patent outputs. PE-backed leveraged buyouts (LBOs) within the UK were found to raise quality-adjusted patent stocks (patent production) by 6% within three years post-acquisition (Amess et al., 2016). Boucly et al. (2011) found a similar trend in French PE-backed LBOs, with patent filings increasing 15% post-investment. Private equity helps reduce financial restrictions and diverts internal resources to focus on long-term R&D projects (Amess et al., 2016, Boucly et al., 2011). Literature on patent output shows mixed results; Lerner et al. (2011) found that the quantity of patent output post-PE investment did not change much, suggesting PE firms boosted patent value, not quantity. Amess (2016) examined the differing impacts of various private equity transaction types on innovation outcomes, finding weak evidence (significant at the 10% level) that public-to-private transactions and divisional buyouts were associated with decreased patent activity (Amess et al., 2016).

Despite the growing body of literature on the effects of PE investment on firm-level innovation, there remains a distinct lack of research specifically focusing on the pharmaceutical sector. Pharmaceutical innovation is unique, with prolonged research cycles, heavy regulatory scrutiny, and high failure rates (Sun et al., 2022). Typical R&D cycles for pharmaceuticals last 10-15 years, while PE investment horizons last 3-7 years (de Concini, 2018). This misalignment between the timelines of the investor and the pharmaceutical innovation cycle creates challenges when measuring innovation. The lack of pharmaceutical-centric research highlights a large void in understanding the relationship between private equity investment and patent production. To understand the influence of private equity firms on pharmaceutical patent production, my initial research question is:

RQ1: To what extent does private equity investment in pharmaceutical firms impact patent outputs?

3.2 Private Equity Impact on the Value of Innovation

Studies on innovation suggest that countries with strong equity markets generate 23% more chemically unique drugs each year, highlighting the importance of developed financial markets for innovation (Benjamin et al., 2022). Developed equity markets provide greater opportunities for

firms to pursue higher-risk R&D projects, as equity financing is uniquely suited for innovation activities since it does not impact firms' financial constraints (Magri, 2014). Lerner et al. (2011) found that patents produced after an LBO received more forward citations on average (a measure of innovation), indicating greater value. Patent citations are official references in patent applications that direct to earlier patents; forward citations are later patents that cite the current patent as a prior reference (Hur and Oh, 2021). Moreno and Epstein (2019) argue the contrary, suggesting that PE firms often implement governance models that may speed up R&D pipelines by prioritizing projects with obvious commercialization pathways, running the risk of ignoring more valuable research and thereby producing lower-quality innovation.

Existing research uses various methods to measure the impact of private equity investment on innovation value. Benjamin et al. (2022) use pharmaceutical exports as a measure of innovation; firms with greater exports are assumed to be more innovative. This metric's flaw is that it does not assess the innovation quality; the new drug may be cheaper. Moren and Epstein (2019) employ value-based pricing frameworks that align drug prices with therapeutic value, arguing that this approach guarantees the reward of innovation. If a drug is higher priced, it is assumed to reflect its innovative impact, not accounting for overpricing by pharmaceuticals. Furthermore, the value-based pricing method does not account for the US's lack of centralized price controls, which could skew valuation methods. Patent-based metrics are often used to measure value, encompassing new developments; most frequently, they refer to forward patent citation rates or economic patent value (Kesselheim and Avorn, 2009).

Glaring gaps are evident by the lack of research focused on the influence PE investment has on pharmaceutical innovation within the US market. The US market is complex, and assigning values to innovation is difficult. Studies within the US regarding the impact of PE on innovation are contradictory; Link and Ruhm (2013) found PE accelerates commercialization in US small business innovation, while Hemphill and Sampat (2012) warn that private equity uses tactics aimed at expanding monopolies without introducing meaningful innovation. The US pharmaceutical industry is unique as it operates without price controls, making it difficult to assign value to innovations (Hemel and Ouellette, 2023). Existing research does not examine the impact of private

equity investment on pharmaceutical innovation value within the US. Due to the existence of these gaps, the following research question is necessary:

RQ2: To what extent does private equity investment impact the value of innovations produced by pharmaceutical firms?

3.3 Private Equity Impact on R&D Expenditure and Financial Health

According to the literature, research and development expenditure of pharmaceuticals is shaped by PE investment in varied degrees. Between 2002 and 2010, Qiu et al. (2014) investigated the role of public and private investment in pharmaceutical R&D expenditure and found that most pharmaceutical R&D is sponsored by private investment. While state funding helped boost less-developed provinces, they contended that PE and venture capital investment most benefited R&D in bigger cities. This emphasizes the vital role private funding has in financing innovation. Su et al. (2021) found that when pharmaceutical firms had high debt levels and ownership concentration, the positive effects of R&D intensity were dampened, arguing that R&D effectiveness is maximized when minimizing debt levels and ownership concentration. R&D intensity is a ratio between a firm's R&D expenditure and total expenditure (Su et al., 2021). PE firms often introduce new debt and concentrate ownership, possibly dampening the effectiveness of R&D expenditure (Tykvová and Borell, 2012).

Recent literature illustrates that PE investment has varying impacts on financial health. Tykvová and Borell (2012) examined the effect of different PE transactions, finding that PE use of leveraged buyouts (LBOs) increased debt levels, limiting the target firm's ability to invest in R&D and mitigate economic downturns. PE-backed firms also exhibited higher leverage ratios and faced greater financial distress when compared to non-PE-backed firms (Tykvová and Borell, 2012). Attig et al. (2012) argue that institutional ownership does not alleviate or worsen financial constraints, reducing financial health. They found that firm-specific factors such as size and industry impacted financial constraint (Attig et al., 2012). Bernstein et al. (2019) found that PE-backed firms in the 2008 financial crisis had 4% more access to debt and 2% more access to equity when compared to peers, allowing them to mitigate financial constraint by improving capital access. Private equity firms have expansive networks that provide companies within their portfolio

greater access to financing in economic downturns, alleviating financial constraints and improving financial health (Bernstein et al., 2019).

Research into the impact of private investment on R&D expenditure in US markets is limited. In the technology sector, PE-backed firms show 12-15% higher R&D intensity than public counterparts, yet non-tech PE-owned firms often prioritize cost-cutting, reducing R&D intensity by 8-10% (Adu-Ameyaw et al., 2022). The literature surrounding the impact of PE investment on R&D expenditure and financial health is mixed. The lack of exploration into the effects of PE investment on US pharmaceuticals' financial health and R&D expenditure has highlighted a gap in existing knowledge; hence, my final research question is:

RQ3: Does private equity investment affect a firm's financial health and spending on research and development?

4. Methodology

This dissertation aims to understand the interplay between private equity investment and innovation outcomes within the pharmaceutical industry in the US. PE's influence on firm performance is well documented in other sectors, yet existing literature fails to explore the impact on innovation metrics within US pharmaceuticals. Understanding the interplay between innovation and PE investment is crucial to understanding whether capital markets are impeding or progressing the development of revolutionary drugs. The analysis is limited to the period between 1985 and 2023 to see the progression of innovation outcomes over an extended period. Patent output and forward patent citation (value) have been taken as the two key measures of innovation. Research and development expenditure and financial health are supplementary metrics that help measure the indirect impacts of private equity investment. The final objective of this dissertation is to improve existing knowledge on the effects of PE investment within the US pharmaceutical industry.

4.1 Hypotheses

All the empirical analysis uses a binary independent variable indicating private equity backing (1) and no private equity backing (0). Before proceeding to regression analysis, t-tests for hypotheses H₁ (innovation output), H₂ (innovation value), H₃ (R&D intensity), and H₅ (financial health) were conducted. These tests compared the means of key variables between PE-backed and non-PE-backed firms, serving to find the baseline variations in the innovation measures, R&D expenditure, and financial health. These t-tests guided model selection and data distribution, flagging potential issues within the dataset.

H₁ **Innovation Output** *Private equity firms boost innovation output*

$$\begin{aligned}
 Innovout_{it} = & \beta_0 + \beta_1 peba_{it} + \beta_2 \ln LP_{it} + \beta_3 \ln zscore_{it} + \beta_4 roe_{it} + \beta_5 gdp_growth_{it} \\
 & + \beta_6 LR_{it} + \beta_7 \ln marcap_{it} + \beta_8 \ln age_{it} + \beta_9 \ln tobin_{it} + \beta_{10} \ln emp_{it} \\
 & + \beta_{11} \ln stow_{it} + \beta_{12} \ln size_{it} + \beta_{13} CF_{it} + \beta_{14} \ln rdint_{it} + \beta_{15} \ln CP_{it} \\
 & + \sum_{p=2}^4 \gamma_p reg_period_p + \sum_{c=2}^4 \delta_c fin_cycle_c + \sum_{e=2}^4 \lambda_e tech_epoch_e + u_i + \varepsilon_{it}
 \end{aligned}$$

A random effect panel regression model was used to better understand the impact of private equity backing on pharmaceutical innovation output. The model uses panel regressions, which control for time-invariant firm-specific factors. This hypothesis aims to expand upon research by Ames et al. (2016), extending to investigate if pharmaceutical innovation output is impacted by private equity investment.

$$\begin{aligned}
 Valinn_{it} = & \beta_0 + \beta_1 peba_{it} + \beta_2 lnLP_{it} + \beta_3 lnzscore_{it} + \beta_4 roe_{it} + \beta_5 gdp_growth_{it} + \beta_6 LR_{it} \\
 & + \beta_7 lnmarcap_{it} + \beta_8 lnage_{it} + \beta_9 lintobin_{it} + \beta_{10} instow_{it} + \beta_{11} lnsiz_{it} \\
 & + \beta_{12} CF_{it} + \beta_{13} lnrdint_{it} + \beta_{14} lnCP_{it} + \sum_{p=2}^4 \gamma_p reg_period_p \\
 & + \sum_{c=2}^4 \delta_c fin_cycle_c + \sum_{e=2}^4 \lambda_e tech_epoch_e + u_i + \varepsilon_{it}
 \end{aligned}$$

Existing literature on private equity's impact on innovation value is mixed; hence, I have run an OLS regression model to determine the effect of private equity backing on pharmaceutical innovation value. The model uses controls for time-invariant and firm-specific factors. This hypothesis aims to expand on Lerner et al. (2011), who used forward patent citations to measure innovation value.

$$\begin{aligned}
 Rdint_{it} = & \beta_0 + \beta_1 peba_{it} + \beta_2 instow_{it} + \beta_3 CF_{it} + \beta_4 lnLP_{it} + \beta_5 gdp_growth_{it} \\
 & + \beta_6 lnzscore_{it} + \beta_7 lnmarcap_{it} + \beta_8 lnCP_{it} + \beta_9 CR_{it} + \beta_{10} lnage_{it} + \beta_{11} roe_{it} \\
 & + \beta_{12} lnsiz_{it} + \beta_{13} lintobin_{it} + \sum_{p=2}^4 \gamma_p reg_period_p + \sum_{c=2}^4 \delta_c fin_cycle_c \\
 & + \sum_{e=2}^4 \lambda_e tech_epoch_e + u_i + \varepsilon_{it}
 \end{aligned}$$

<i>H₄</i>	<i>Leverage Impact on R&D Intensity</i>	<i>Private equity investment decreases leverage's impact on R&D intensity</i>
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$$\begin{aligned}
Rdint_{it} = & \beta_0 + \beta_1 peba_{it} + \beta_2 pebalr_{it} + \beta_3 instow_{it} + \beta_4 CF_{it} + \beta_5 lnLP_{it} + \beta_6 gdp_growth_{it} \\
& + \beta_7 lnzscore_{it} + \beta_8 lnmarcap_{it} + \beta_9 lnCP_{it} + \beta_{10} CR_{it} + \beta_{11} lnage_{it} \\
& + \beta_{12} roe_{it} + \beta_{13} lnsize_{it} + \beta_{14} lntobin_{it} + \sum_{p=2}^4 \gamma_p reg_period_p \\
& + \sum_{c=2}^4 \delta_c fin_cycle_c + \sum_{e=2}^4 \lambda_e tech_epoch_e + u_i + \varepsilon_{it}
\end{aligned}$$

To understand the impact of PE investment on research and development (R&D) expenditure, I performed random-effect panel regressions that compared PE-backed and non-PE-backed pharmaceutical firms. The literature highlighted differing perspectives on the impact of PE backing on R&D expenditure; hence, hypothesis three explores the direct impact on R&D expenditure relative to total sales (R&D intensity). Tykvová and Borell (2012) highlighted that the increased debt brought by private equity dampened R&D effectiveness; hence, hypothesis four aims to understand if PE-backing improves or mitigates the adverse effects of leverage on R&D expenditure within pharmaceuticals. Hypothesis four also aims to investigate if Su et al. (2021) findings that firms with high R&D intensity and low debt perform better than their competitors are true for the pharmaceutical industry. Hypothesis four extends model three, adding an interaction term to account for financial leverage.

$$\begin{aligned}
Zscore_{it} = & \beta_0 + \beta_1 peba_{it} + \beta_2 instow_{it} + \beta_3 CF_{it} + \beta_4 lnrdint_{it} + \beta_5 lnLP_{it} \\
& + \beta_6 gdp_growth_{it} + \beta_7 lnmarcap_{it} + \beta_8 lnCP_{it} + \beta_9 lnage_{it} + \beta_{10} roe_{it} \\
& + \beta_{11} LR_{it} + \beta_{12} lnsize_{it} + \beta_{13} lntobin_{it} + \sum_{p=2}^4 \gamma_p reg_period_p \\
& + \sum_{c=2}^4 \delta_c fin_cycle_c + \sum_{e=2}^4 \lambda_e tech_epoch_e + u_i + \varepsilon_{it}
\end{aligned}$$

Financial health was consistently mentioned in the literature as a determinant of innovation in firms after receiving PE investment. Hypothesis five aims to investigate the impact of private equity investment on financial health within pharmaceutical firms. The model uses a random effect panel regression, taking Altman's Z-score as the measure of financial health. This model aims to investigate whether Su et al. (2021)'s findings that PE firms increase financial constraints on firms, reducing financial health, are consistent within the pharmaceutical sector.

4.2 Data Collection

Five different databases comprise the dataset developed for the empirical research of this dissertation. I combined every database using 9-digit CUSIP or PERMNO codes, a unique identification code used to identify financial securities and firms (Booth, 2025). Data from every database was collected between 1985 and 2023. All US pharmaceutical companies were compiled from the Refinitiv Eikon database along with their PE-backed status, past, present, or none. The NAICS classification helped filter the data for pharmaceutical firms. My initial dataset consisted of 899 companies; however, data merging reduced it to 394.

I used the North America Capital IQ's Compustat database on WRDS to get my accounting data; it provides company-specific basic annual data and financial ratios. When gathering data from Compustat, I did not use filters since I often noticed mismatched NAICS codes when merging with other databases. I gathered historic market data using the CRSP/Compustat merged database on

WRDS. This database links the CRSP database with Compustat, providing information on corporate actions and past historical market data.

The Sero (2023) study compiled citation data for all US companies and patent output data. The methodology they developed calculates the economic importance of each innovation, estimating the value of each patent created and matching them with company PERMNO codes. Papanikolaou and Seru created the 2023 database, extending their original findings from 2017 to 2023 (Kogan et al., 2017). The Federal Reserve Economic Database was used to collect information on annual US GDP growth.

4.3 Variable Selection

Table 1: Independent Variable and Descriptive Statistics

Variable	Form	Mean	Std. Dev.	Min	Max
Private Equity Backed Indicator	Binary	0.5347	0.4988	0	1

The PE-backed indicator variable indicates whether a firm has received support from a private equity firm in its history. It does not discriminate on PE investment length or timing; instead, it just accounts for whether a pharmaceutical firm has ever received PE investment. The mean is above 0.5, indicating that more firms in the data set were PE-backed.

Table 2: Dependent Variable Descriptive Statistics

Variable	Form	Mean	Std. Dev.	Min	Max
Innovation Output	LN	0.5912	0.9493	0.0000	3.1355
Value of Innovation Output	LN	0.4674	0.9818	0.0000	3.2958
R&D Intensity	LN	-1.6193	1.2997	-4.3472	0.5114
Z-Score	LN	0.3134	0.9547	-6.2927	2.1920

LN transformation indicates natural logarithmic; Level indicates the variable was not transformed.

Each dependent variable was carefully created and adjusted to answer the hypotheses (Appendix A). The standard deviations of the four dependent variables are relatively large compared to their

means, indicating substantial heterogeneity in the metrics across firms (Table 2). This is standard in most innovation data. Following Allen et al. (2022), annual firm-level patent counts were used to estimate innovation output (H1), with the variable winsorized at both tails of the 5th percentile level to reduce the impact of extreme outliers and log-transformed to handle skewness. Following Katila (2000), average forward patent citations per firm were used, winsorizing at both tails of the 5th percentile level and log-transforming to standardize the skewed distribution, estimating the value of innovation output (H2).

The ratio of R&D expenses to total expenditure, R&D intensity (H3), was used to measure firms' R&D expenditure, log-transforming and winsorizing at both tails of the 5th percentile level to minimize the impact of scale and outliers. An interaction term between leverage ratio and PE backing was included to assess the moderating effect of debt (H4), looking at whether greater debt levels change the influence of private equity on R&D intensity. Altman's Z-score combined several balance sheet and income statement measures measuring financial health (H5); this variable was logarithmically transformed and winsorized at both tails of the 5th percentile level to lessen the impact of outliers. These adjustments were selected to solve skewness, guarantee comparability between companies, and fit accepted approaches in the PE literature on innovation.

Table 3: Control Variable Transformations and Descriptive Statistics

Control Variable	Transformation	Mean	Std. Dev.	Min	Max
Labor Productivity	LN	4.8892	1.6276	-4.1092	7.0571
Z Score	LN	0.3134	0.9547	-6.2927	2.1920
Return on Equity	Level	-0.4661	1.3366	-3.9336	2.4651
GDP Growth	Level	5.0661	2.7523	-0.7261	12.2720
Leverage Ratio	Level	0.5575	0.6141	0.0524	2.5825
Financial Market Development	LN	1.8530	1.4662	-0.7062	4.4293
Age	LN	0.9961	0.2950	0.3266	1.3368
Tobin Q	LN	0.9497	0.6797	-0.2575	2.2057
Employee Count	LN	-2.3663	2.1050	-5.5215	2.8904
Institutional Ownership	Level	0.4458	0.3419	0.0018	1.0109
Size	LN	4.0946	2.3256	-0.1054	9.1685
Capital Ratio	Level	0.4357	0.6215	-1.6169	0.9470
Cash Flow	Level	-0.4329	0.6303	-2.3392	0.2077
R&D Intensity	LN	-1.6193	1.2997	-4.3472	0.5114
Capital Productivity	LN	1.0904	1.8074	-9.6970	3.9443

LN indicates natural logarithmic transformation; Level indicates no transformation.

Established literature guided the selection and calculation of control variables, which include macroeconomic conditions, firm characteristics, and governance structures (Appendix B). Skewness and scale were mitigated for some control variables by natural logarithmic transformations. Labor productivity, drawn from Ames et al. (2016), shows the company's operational efficiency relative to production. Altman's Z-score captures financial health, as Tykvová and Borell (2012) advised, to evaluate distress in risks in private equity-backed companies.

Based on Yang et al. (2024), return on equity captures profitability. While macroeconomic conditions were controlled by GDP growth, leverage ratio captured the capital structure dynamics, aligning with Tykvová and Borell (2012). Firm maturity proxied by age and institutional ownership was included to account for lifetime and governance effects as in Coas et al. (2018) and Armitage et al. (2025). Market capitalization is extensively used in the literature to measure financial market

development, capturing the depth of financial markets (Svirydzenka, 2016). Tobin Q controls for firm value and the growth opportunities a firm has over time (Meles et al., 2014).

The Pearson correlation matrix shows low-to-moderate correlations (Appendix C). Size and employee count, both measures of scale, showed a strong correlation ($p=0.85$); hence, they will not be used in the same model together. Capital and leverage ratios, reflecting their complementary functions in capital structure, showed an inverse relationship ($p=-0.77$). Most correlations were less than 0.6, indicating limited multicollinearity issues.

Table 4: Fixed Effects

Fixed Effects	Variable Acronym
Financial Cycle	Fin_cycle
Technological Developments	Tech_epoch
Regulatory Period	Reg_period

Time-fixed effects helped the models solve possible multicollinearity problems. The financial cycle variable is adapted from Lichtenberg's (2018) research into the impact of new drug developments. The financial cycle time-fixed effect takes four separate financial periods: 1985–2000, low-interest R&D funding; 2001–2008, surge in biotech investment following the dot-com era; 2009–2015, a quantitative easing period; and 2016–2023, the SPAC/IPO boom and rising interest rates. Inspired by Peters (2009), who examined the consistency of innovation, the technological development variable considers the recombinant DNA era spanning 1985 to 1994; the genomics revolution spanning 1995 to 2004; the evolution of targeted therapies spanning 2005 to 2014; and from 2015 to 2023, AI and machine learning-driven drug discoveries. The regulatory period variable is adapted from Lichtenberg (2018), considering four main periods: 1985–1996, the pre-FDA Modernization Act; 1997–2009, the Prescription Drug User Fee Act; 2010–2014, the Affordable Care Act; and 2015–2023, the therapy designation era.

5. Empirical Analysis

5.1 T-Test and Graphs

Significant gaps were present in the datasets during the data collection process, and some data was lost during the merging process; hence, each variable has varying numbers of observations. In the final sample, I had 394 pharmaceutical firms, 202 of which were currently or had been PE-backed. The precision of my estimates will vary, as the models with fewer observations will possess weaker statistical power.

Each variable is reported for two groups: PE-backed and non-PE-backed. The two-tailed t-tests comparing the sample means of the two data subsets indicate that the means are statistically different for innovation output, innovation value, R&D intensity, and financial health, justifying the pursuit of regression analysis (Appendix D). All four had significance values below the 1% level and large t-stats (Table 6). The t-tests provide glaring insights into the impact of private equity investment on pharmaceuticals. PE-backed firms had a larger mean (0.7630), indicating greater innovation output when compared to non-PE-backed firms (0.3953). However, PE-backed firms had a higher standard deviation, indicating greater volatility in innovation output. A similar trend can be observed for innovation value, with the PE-backed pharmaceuticals mean double (0.6064) that of non-PE-backed (0.3089)

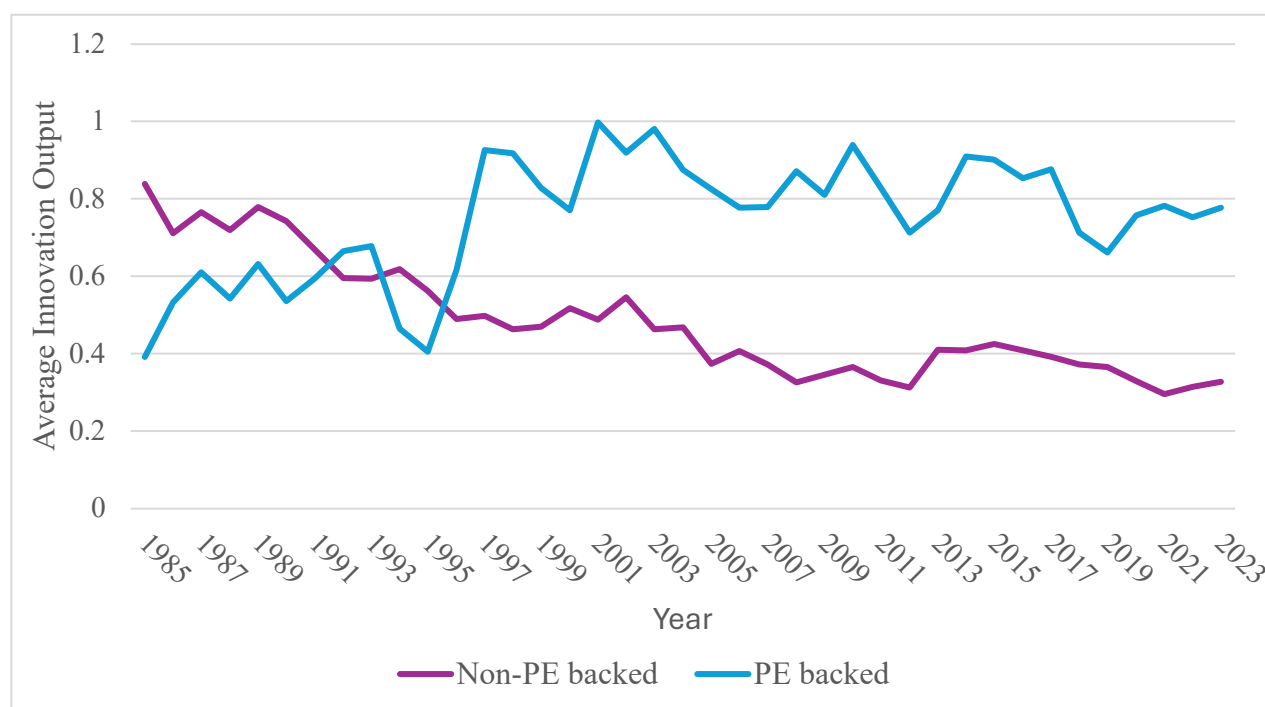
Firms with PE-backing have greater R&D intensity, with a less negative logarithmic mean (-1.4169) than non-PE-backed firms (-1.8755). The lower standard deviation (1.1128) for PE-backed firms signifies more consistency in their R&D spending. Moreover, PE-backed pharmaceuticals exhibited a lower z-score (0.2781), hence worse financial health when compared to non-PE-backed firms (0.3538). The results of these t-tests do not provide insight into why PE-backed firms have worse financial health.

Table 5: Descriptive Statistics for Private Equity Backing

Private Equity Backing		Obs	Mean	Std. Err.	Std. Dev.	T-Stat	p-value
Innovation Output	1: PE Backed	4,048	0.7630	0.0156	0.9916	-17.17	<0.001
	0: No PE Backing	3,551	0.3953	0.0144	0.8578	-17.17	<0.001
Innovation Value	1: PE Backed	4,048	0.6064	0.0169	1.0781	-13.33	<0.001
	0: No PE Backing	3,551	0.3089	0.0139	0.8312	-13.33	<0.001
R&D Intensity	1: PE Backed	3,919	-1.4169	0.0178	1.1128	-13.77	<0.001
	0: No PE Backing	3,095	-1.8755	0.0263	1.4640	-13.77	<0.001
Z-Score	1: PE Backed	4,048	0.2781	0.0151	0.9658	3.45	<0.001
	0: No PE Backing	3,551	0.3538	0.0158	0.9404	3.45	<0.001

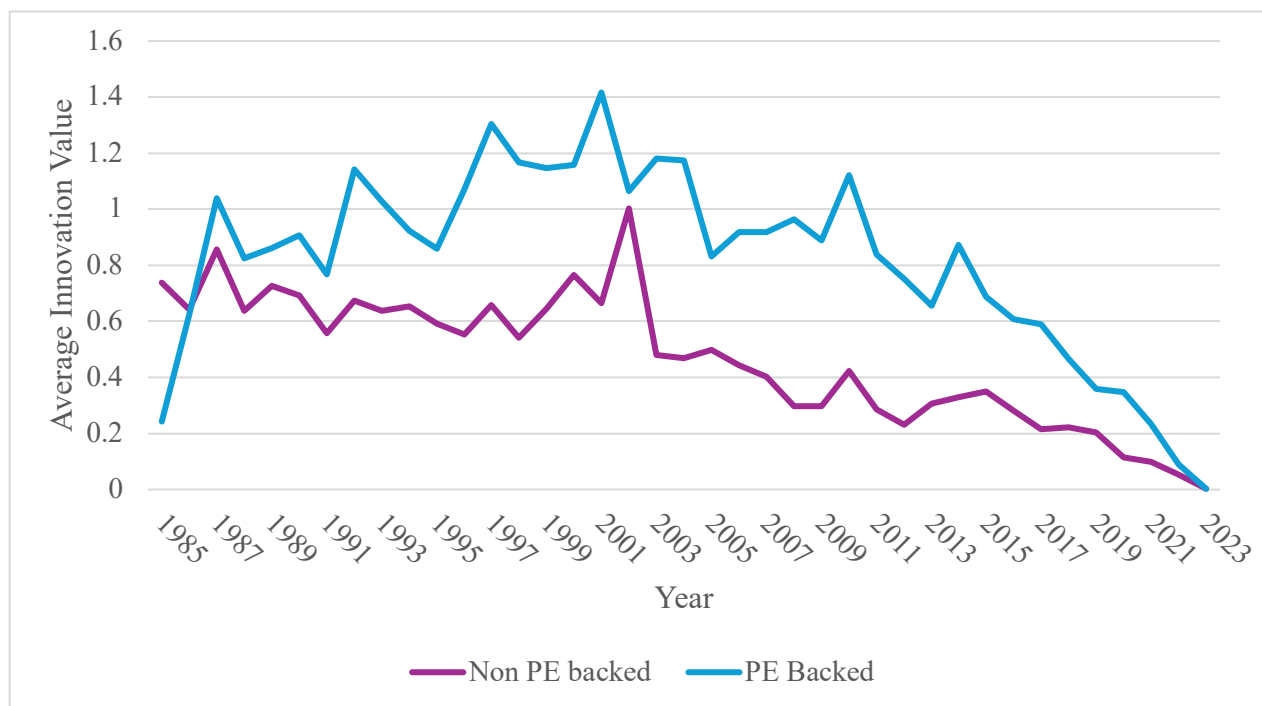
Graphically representing these relationships exposes further insights. The data in each graph was aggregated by PE-backed status and year, providing the average for each year. Graphically plotting innovation output highlights that PE-backed pharmaceuticals have outperformed non-PE-backed pharmaceuticals (Graph 3). Between 1985 and 1995, innovation output for non-PE-backed pharmaceuticals was greater, yet a consistent downward trend has put them behind PE-backed pharmaceuticals since the mid-1990s. On the other hand, PE-backed pharmaceuticals had an explosion of innovation output in the mid-1990s, which has also gradually decreased over time. As noted in the literature, the swift rise in innovation output coincided with heightened private equity interest in the pharmaceutical sector (Busfield, 2020).

Graph 3: Average Innovation Output by Private Equity Status over Time



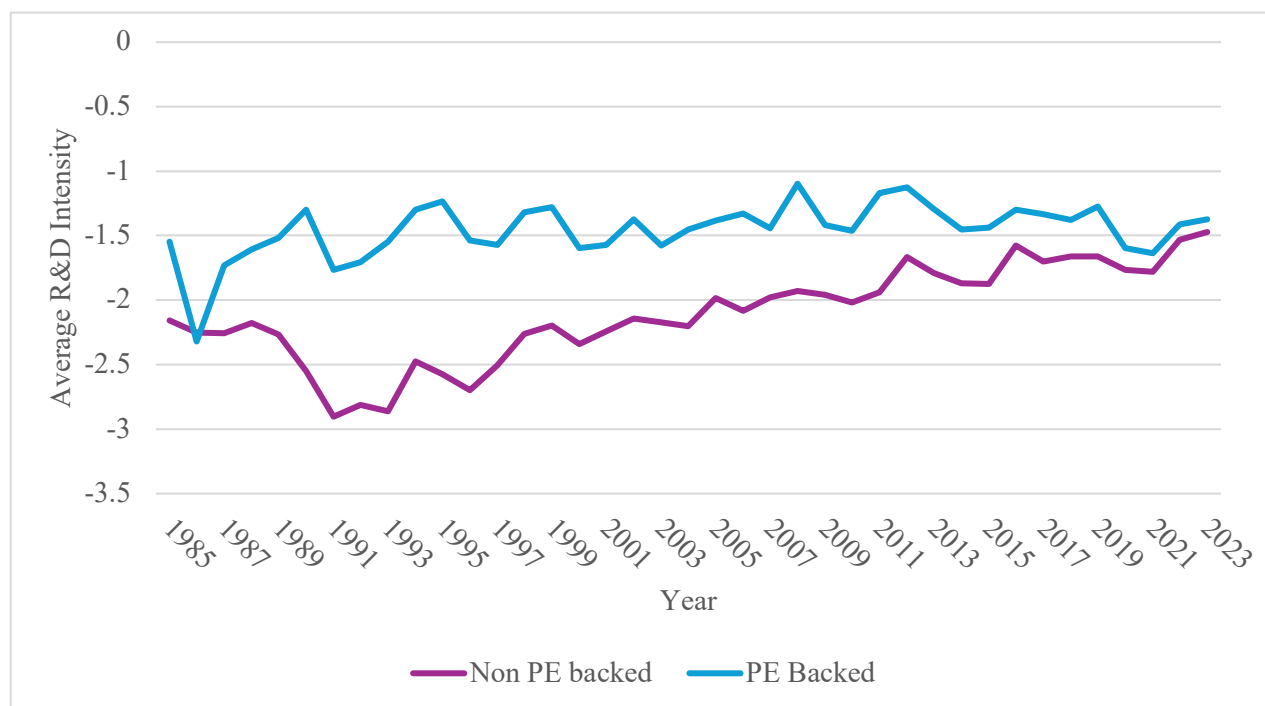
Innovation value follows a different trend from innovation output. PE-backed firms experienced a rapid increase in innovation value in the late 1980s, which continued to increase into the early 2000s and began to decline (Graph 4). It is consistently observed that non-PE-backed firms have lower innovation value, with a consistent decline in innovation value until 2023. The trend of this graph is likely not illustrating the whole picture. The method for calculating innovation value uses forward citations, which are biased by time. It relies on future references to a new patent; hence, it is unlikely that a patent filed in 2023 would have many future citations, thus the sharp downward trajectory.

Graph 4: Average Value of Innovation Output by PE Status over Time

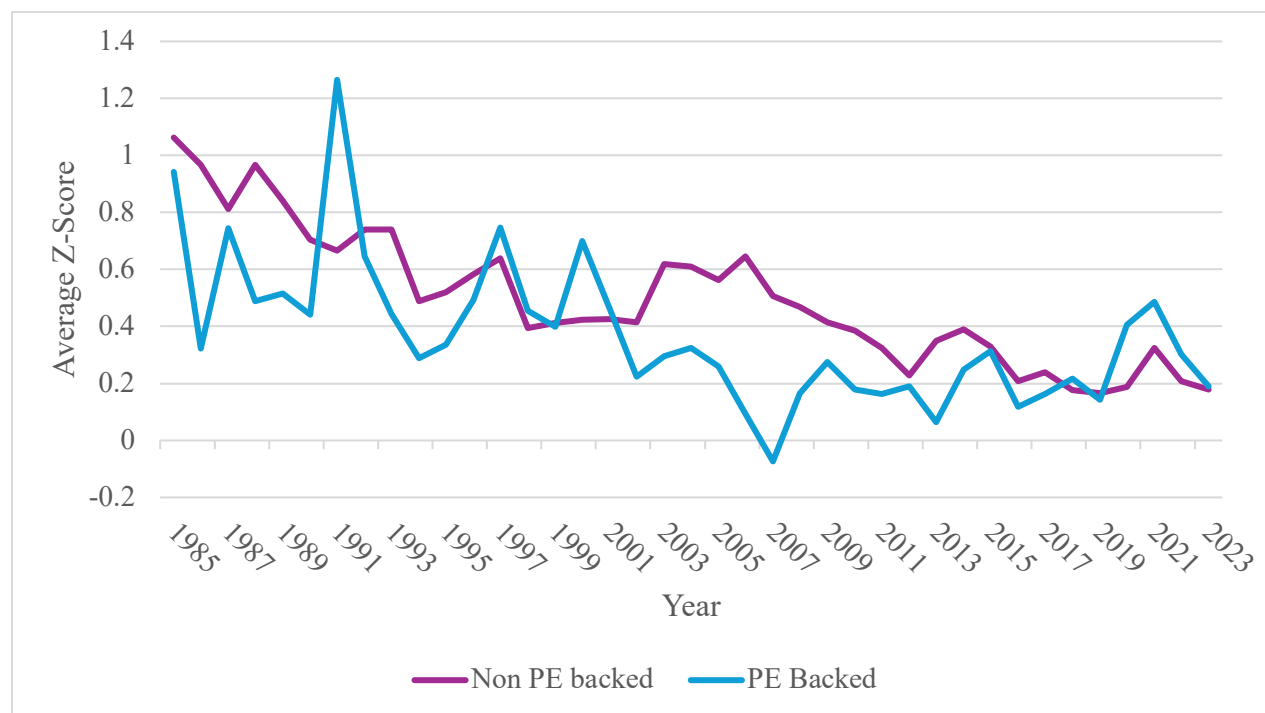


Graphical trends show that PE-backed pharmaceuticals have continuously exhibited greater research and development activity (Graph 5). Although PE-backed pharmaceuticals' R&D intensity has stagnated at a similar level, non-PE-backed pharmaceuticals have gradually risen over the past four decades. Interestingly, financial health across PE-backed and non-PE-backed pharmaceuticals has wildly fluctuated over time (Graph 6). PE-backed and non-PE-backed pharmaceuticals follow a remarkably close and similar trend, diverging very little. Notably, PE-backed pharmaceuticals in the 2000s saw worse financial health, presumably due to the higher leverage often plaguing PE-backed firms (Axelson et al., 2013). Pharmaceuticals with better financial health have a greater ability to increase R&D expenditure. Non-PE backed firms seem to have less variance in financial health, possibly indicating PE-backed firms face greater financial health volatility.

Graph 5: Average R&D Intensity by PE Status over Time



Graph 6: Z-score (Financial Health) by PE Status over Time



All the graphs together highlight an interesting trend across PE and non-PE-backed firms. Innovation values are declining, yet R&D intensity has remained stagnant or increased. Innovation outputs have remained stagnant, slightly decreasing since the late 1990s. The financial health of pharmaceuticals worsened in the post-2008 crisis; however, it has rebounded again to pre-financial crisis levels. This highlights a slowdown in innovation in the pharmaceutical industry. This could be caused by most pharmaceutical innovation originating from incremental improvements or “me-too” drugs rather than actual breakthroughs, as highlighted by (Fernald et al., 2024).

5.3 Regression Models

Random effect panel regressions were conducted to evaluate four hypotheses, with the value of the innovation model using an OLS regression. All the models used robust standard errors and featured fixed time variables to control for period-specific effects. When analyzing coefficients, all the logarithmically transformed variables had their coefficients exponentiated to provide the multiplicative effect, which converts the coefficient into an interpretable percentage. If the variable was not logarithmically transformed, it was interpreted as is.

Model 1:*Table 6: Random Effect Panel Regression of Innovation Output with Robust Standard Errors*

Variable	Coefficient	Estimated % Change in Innovation Output	Significance
PE Backed Status	0.2840	32.84%	***
Labor Productivity	0.0010	0.10%	
Z Score	-0.0380	-3.73%	*
Return on Equity	-0.0090	-0.90%	
GDP Growth	0.0010	0.10%	
Leverage Ratio	0.0720	7.20%	
Financial Market Development	-0.0170	-1.69%	
Age	0.7680	115.55%	***
Tobin Q	0.0600	6.18%	
Employee Count	0.0690	7.14%	*
Institutional Ownership	0.0970	10.19%	
Size	0.1650	17.94%	***
Cash Flow	-0.0760	-7.32%	
R&D Intensity	0.0960	10.08%	***
Capital Productivity	0.0048	0.48%	

Significance: * indicates p-value<0.1, ** indicates p-value<0.05, *** indicates p-value<0.01

This model aimed to investigate the impact of PE-backing on a firm's innovation output (Table 6, Appendix F). The model indicated that PE-backing boosted innovation output when controlling for firm effects and fixed-time effects, aligning with hypothesis one. The model suggests PE-backed pharmaceuticals increase innovative output by 32.84% compared to non-PE-backed firms, at the 1% significance level, controlling for other firm characteristics and time fixed effects. This model aligns with the t-tests and graphical analysis, indicating hypothesis one has statistical backing.

Some of the standardized control variables in the model were statistically significant. Both firm age and size are directly linked to boosting firm innovation output, with a unit increase in each leading to a 115.55% and 17.94% increase in innovation output, respectively. This highlights that older and larger firms produce more innovations, possibly due to economies of scale or having

existing profit streams. A unit increase in R&D intensity leads to a 10.08% increase in innovation output. This verifies the direct link between R&D expenditure and the production of innovation.

Model 2

Table 7: OLS Regression of Value of Innovation Output with Robust Standard Errors

Variable	Coefficient	Estimated % Change in Value of Innovation	Significance
PE Backing	0.2560	29.18%	***
Labor Productivity	-0.0250	-2.47%	
Z-Score	-0.0800	-7.7%	***
Return on Equity	-0.0140	-1.40%	
GDP Growth	-0.0160	-1.60%	***
Leverage Ratio	-0.0360	-3.60%	
Financial Market Development	0.0040	0.40%	
Age	0.2940	34.18%	**
Tobin-Q	0.0880	9.20%	
Institutional Ownership	0.1040	10.40%	
Size	0.1160	12.30%	***
Cash Flow	0.0060	0.60%	
R&D Intensity	0.1170	12.41%	***
Capital Productivity	0.0140	1.41%	

Significance: * indicates p-value<0.1, ** indicates p-value<0.05, *** indicates p-value<0.01

This model aimed to investigate the impact of PE-backing on the value of innovation output. The pooled OLS regression model indicates PE-backing has a highly statistically significant effect on innovation output, increasing innovation value by 29.18% compared to non-PE-backed firms (Table 7, Appendix H). The results of this model align with the t-tests and graphical analysis, highlighting the positive impact of private equity in pharmaceuticals, aligning with hypothesis two. Interestingly, a unit increase in GDP growth is associated with a -1.6% decrease in innovation value. This may seem counterintuitive; however, it may be that the pharmaceutical companies divert resources towards established products over new innovative drugs, or the increased market stability may reduce pharmaceutical interest in pursuing innovative drugs.

In line with the existing literature, a unit increase in R&D intensity leads to a 12.41% increase in innovation value. This highlights the direct link between R&D expenditure and innovation value; pharmaceuticals dedicating resources to high R&D projects are likely to produce more innovative drugs. Furthermore, this model highlights that a unit increase in firm size leads to a 12.30% increase in innovation value, which may link larger pharmaceuticals with producing more innovative drugs.

Model 3

Table 8: Random Effect Panel Regression of R&D Intensity with Robust Standard Errors

Variable	Coefficient	Estimated % Change in R&D Intensity	Significance
PE Backed Status	0.5470	72.81%	***
Institutional Ownership	0.0190	1.90%	
Cash Flow	-0.9010	-90.10%	***
Labor Productivity	0.0360	3.67%	
GDP Growth	-0.0002	-0.02%	
Z-Score	-0.0370	-3.63%	*
Financial Market Development	-0.1130	-10.68%	***
Capital Productivity	-0.0450	-4.40%	
Capital Ratio	-0.0720	-7.20%	
Age	-0.3880	-32.16%	**
Return on Equity	-0.0510	-5.10%	***
Size	-0.0740	-7.13%	*
Tobin-Q	0.2780	32.05%	***

Significance: * indicates p-value<0.1, ** indicates p-value<0.05, *** indicates p-value<0.01

This model aimed to investigate the impact of PE on R&D intensity (R&D expenditure). A random effect panel regression yielded statistically significant results at the 1% level, indicating that PE-backed firms had 72.81% greater R&D intensity than non-PE-backed firms, controlling for other factors, aligning with hypothesis three (Table 8, Appendix J). This aligns with the literature, as PE firms often enter firms with promising growth opportunities, and often in pharmaceuticals, these opportunities are only obtainable via greater expenditure into R&D. Interestingly a unit increase

in cash flows yields to a -90.1% decrease in R&D intensity, highlighting the inverse relationship between firm cash flow and R&D expenditure. A unit increase in Tobin-Q (growth opportunities) results in a 32.05% increase in R&D intensity. This suggests that firms that pursue greater R&D expenditure have greater growth opportunities and better market sentiment.

Higher ROEs were associated with lower R&D intensity, with a unit increase in ROE resulting in a 5.1% decrease in R&D intensity. This highlights that firms focusing on high ROE exploit profitable operations rather than invest in uncertain R&D projects. This aligns with the literature, which discusses the current prioritization of “me-too” drugs rather than innovative new solutions within the pharmaceutical industry (Aronson, 2020). Interestingly, a unit increase in financial market development was associated with a 10.68% decrease in R&D intensity. This could highlight that pharmaceuticals in more developed financial markets may prefer a financialized business model based on existing products, putting less priority on R&D.

Model 4

Table 9: Random Effect Panel Regression of R&D Intensity with Financial Leverage Interaction Term with Robust Standard Errors

Variable	Coefficient	Estimated % Change in R&D intensity	Significance
PE Backed Status	0.5980	81.8%	***
PE-Backed Interaction	-0.1130	-10.7%	
Institutional Ownership	0.0200	2.00%	
Cash Flow	-0.9030	-90.30%	***
Labor Productivity	0.0370	3.77%	
GDP Growth	-0.0001	-0.01%	
Z-Score	-0.0380	-3.7%	**
Financial Market Development	-0.1130	-10.68%	***
Capital Productivity	-0.0440	-4.30%	
Capital Ratio	-0.1410	-14.10%	
Age	-0.3860	-32.02%	**
Return on Equity	-0.0500	-5.00%	***
Size	-0.0740	-7.13%	*
Tobin-Q	0.2780	32.05%	***

Significance: * indicates p-value<0.1, ** indicates p-value<0.05, *** indicates p-value<0.01

This model aimed to investigate the impact of increased leverage on firms' R&D intensity (Table 9, Appendix L). The model yielded statistically insignificant results for the interaction term between financial leverage and private equity backing, suggesting the data does not support the claim of hypothesis four. These results indicate that PE-backings' impact on financial leverage does not dampen firm R&D intensity. Similarly to the previous regression model, the same control variables indicated statistical significance.

Model 5

Table 10: Random Effect Panel Regression of Financial Constraint with Robust Standard Errors

Coefficient	Significance	Estimated % Change in Z- Score	Significance
PE Backed Status	-0.0640	-6.2%	
Institutional Ownership	-0.2020	-20.2%	*
Cash Flow	0.0630	6.3%	
R&D Intensity	-0.0860	-8.2%	***
Labor Productivity	-0.0610	-5.9%	**
GDP Growth	-0.0050	-0.5%	
Financial Market Development	0.0890	9.3%	***
Capital Productivity	0.0660	6.8%	***
Age	-0.0390	-3.8%	
Return on Equity	0.0630	6.3%	***
Leverage Ratio	-0.6740	-67.4%	***
Size	0.0630	6.5%	**
Tobin-Q	0.0140	1.4%	

Significance: * indicates p-value<0.1, ** indicates p-value<0.05, *** indicates p-value<0.01

This model investigated the impacts of PE-backing on pharmaceuticals' financial health. The random effect panel regression yielded statistically insignificant results, indicating that PE backing does not affect pharmaceuticals' financial health (Table 10, Appendix N). This result differs from the t-tests and graphical analysis. Interestingly, a unit increase in R&D intensity was associated with an 8.2 decrease in financial health. This highlights the strain R&D investment has on pharmaceutical firms, possibly highlighting why pharmaceuticals may avoid pursuing risky new drugs.

As expected, a unit increase in leverage ratio was associated with a 67.4% decrease in financial health, highlighting the detrimental impact of debt on pharmaceutical financial health. Firms in more developed financial markets saw a 9.3% increase in financial health for a one-unit increase in financial market development. This highlights that firms in developed financial markets have better access to capital and greater flexibility. Notably, a unit increase in capital productivity led

to a 6.8% increase in financial health. Firms with more effective capital have higher returns, hence face less financial constraint, improving financial health.

5.4 Robustness Check

5.4a Hausman Test

Table 11: Hausman Test

Variable	X² Stat	p-Value	Model Preference
Innovation Output	40.97	0.004	Fixed Effects
Innovation Value	56.71	0.000	Fixed Effects
R&D Intensity	247.09	0.000	Fixed Effects
R&D Intensity with Interaction	249.89	0.000	Fixed Effects

All the models used panel regressions, and to decide whether to use fixed or random effects, I used Hausman's Test. The Hausman tests indicated a preference for fixed effects with p-stats below the 5% level. However, my independent variable was excluded from the fixed effect models due to its binary nature; hence, random effects were used despite being the inferior choice (Table 11).

5.4b Assumptions Testing

Various statistical tests were used to ensure the robustness of my regressions. Value inflation factors were used to detect multicollinearity among variables, with a VIF above 5 indicating high multicollinearity. All the models featured most of the control variables specified; variables were only omitted if they had a high value inflation factor (VIF) and were redundant. If two variables measured similar metrics, the variable with a higher VIF was omitted.

The Wooldridge test for autocorrelation in panel data was used to detect whether residuals are correlated with firms over time. The test indicated the presence of autocorrelation with a significant p-stat, hence clustered robust standard errors were used in the models (Appendix O). The Breusch-Pagan Lagrange multiplier test for random effects was used to test whether a random effect model is appropriate compared to pooled OLS. The test indicated that only my innovation value model required using pooled OLS, since it had a p-stat above 0.05 (Appendix P).

The Shapiro-Wilk test for normality helped test the normality of the predicted random effects. The test indicated that not all the models were normally distributed (Appendix Q). The clustered robust standard errors do not rely on the normality assumption, improving the reliability of my statistical inferences. Furthermore, many of the variables were transformed to help with this assumption. Using robust standard errors also helps mitigate the effects of heteroskedasticity; I could not find a direct test for heteroskedasticity in the random effect models. Hence, robust standard errors were used to mitigate the impact of model misspecification.

5.4c Propensity Score Matching

PE firms often are selective about which firms they invest in; hence, as a method to prevent selection bias, propensity score matching (PSM) was employed as a robustness check. Firms were matched within years using a composite score of control variables, combining the estimated propensity scores and year indicators, helping ensure comparability. This separated the firms into matched and unmatched. Only matched firms were kept in the PSM models, and unmatched groups were removed (Appendix U, V, W, X, Y). After treating the variables, the mean bias across all variables went from 12.3 to 3.9, and the median bias went from 4.5 to 2.1 (Appendix R, S). The model before had a B-statistic (multivariate bias) of 96.1*, which indicated severe bias present, to 15.9, an acceptable level of bias in the model (Appendix R, S).

The propensity score matching has improved the comparability of control variables, improving the explanatory power of my regression models. When rerunning the models, I found the PSM regressions provided more substantial evidence for the positive impact of PE backing on innovation output and R&D intensity (Appendix U, W). For instance, innovation output was found to have a larger effect size of 35.39% in the PSM model compared to 32.84% in the non-PSM model (Appendix U, Table 6). On the other hand, innovation value had a smaller effect size of 24.23% in the PSM model compared to the non-PSM model, 29.18% (Appendix V, Table 7). PSM helps improve the significance of variables, reduces variability, and makes the control variables behave more predictably, allowing for better empirical analysis.

6. Conclusion

The findings of this report create a comprehensive empirical assessment of the impact of private equity investment on innovation within the US pharmaceutical industry. Four decades of firm-level data and robust statistical analysis provide a nuanced insight into how PE involvement impacts innovation, uncovering the benefits and inherent limitations. Two-tailed t-tests indicated PE-backed firms tend to exhibit higher R&D intensity, greater innovation output, and value to non-PE-backed firms. The t-tests indicated PE-backed pharmaceuticals had worse financial health. Graphical analysis revealed an industry-wide stagnation or decline in the innovation metrics. This reflects the industry trends of increased development costs and a shift toward incremental drugs rather than breakthrough innovations.

The regression models controlling numerous factors indicated that PE backing is linked with a statistically significant increase in innovation output. This finding aligns with Ames et al. (2016), who found similar UK PE investment trends. The second model highlighted that PE-backing boosted innovation value, aligning with the findings of Lerner et al. (2011), who found PE-backed firms received more forward citations on average. Curiously, R&D intensity was boosted in pharmaceuticals with PE-backing, disagreeing with the findings of Adu-Ameyaw et al. (2022), who saw an 8-10% decrease in R&D intensity in non-tech firms. The literature often links PE-backing with dampening effects on R&D expenditure caused by elevated leverage, as in Tyková and Borell (2012). By contrast, Model 4 found no statistically significant correlation between PE-backing and leverage. Moreover, Model 5 found no statistically significant linkage between PE-backing and financial health, agreeing with the findings of Attig et al. (2012) and disagreeing with Su et al. (2021).

There are numerous methodological issues present in my empirical analysis. All my models possessed issues with autocorrelation, non-normality, and heteroskedasticity. These errors likely originated from issues within the dataset, as I did not account for PE firms' entry and exit periods, hindering my ability to isolate firm-specific effects. The dataset also faced several problems with data quality due to missing data points. My model was mis-specified since I couldn't use fixed effects for my regression analysis. Furthermore, Abrams and Sampat (2019) found a weak

relationship between citations and therapeutic value in pharmaceuticals, highlighting a possible weakness in taking forward citations as a measure of innovation. Firm-specific data for other innovation metrics was challenging to collect; hence, I used forward citations.

In conclusion, this report advances the understanding of private equity's impact within the pharmaceutical industry. It emphasizes private equity's positive impact on pharmaceuticals, boosting R&D expenditure and increasing innovation output and value. My empirical analysis leaves room for further investigations. Future studies should investigate the processes employed by PE to influence innovation, such as changes in management, rehauling of the R&D process, and capital allocation strategies. Micro-level analysis of pharmaceuticals would help indicate which of PE's direct activities fosters more innovation.

7. References

- ADU-AMEYAW, E., DANSO, A., HICKSON, L. & LARTEY, T. 2022. R&D spending intensity of private vs public firms: the role of cash flow, leverage and information quality. *Journal of applied accounting research*, 23, 770-787.
- AITKEN, M., KLEINROCK, M. & PRITCHETT, J. 2022. *Emerging Biopharma's Contribution to Innovation*, IQVIA Institute For Human Data Science.
- AMESS, K., STIEBALE, J. & WRIGHT, M. 2016. The impact of private equity on firms' patenting activity. *European economic review*, 86, 147-160.
- ANDERSSON, T., GLEADLE, P., HASLAM, C. & TSITSIANIS, N. 2010. Bio-pharma: A financialized business model. *Critical Perspectives on Accounting*, 21, 631-641.
- ANGELL, M. & ANGELL, M. 2005. *The truth about the drug companies: How they deceive us and what to do about it*, Random House Trade Paperbacks New York.
- ATTIG, N., CLEARY, S., EL GHOUL, S. & GUEDHAMI, O. 2012. Institutional investment horizon and investment–cash flow sensitivity. *Journal of banking & finance*, 36, 1164-1180.
- AXELSON, U. L. F., JENKINSON, T. I. M., STRÖMBERG, P. E. R. & WEISBACH, M. S. 2013. Borrow Cheap, Buy High? The Determinants of Leverage and Pricing in Buyouts. *The Journal of finance (New York)*, 68, 2223-2267.
- AZOULAY, P., ZIVIN, J. S. G., LI, D. & SAMPAT, B. N. 2019. Public R&D Investments and Private-sector Patenting: Evidence from NIH Funding Rules. *The Review of economic studies*, 86, 117-152.
- BABUS, A. & PARLATORE, C. 2022. Strategic fragmented markets. *Journal of Financial Economics*, 145, 876-908.
- BENJAMIN, M. B., TODD, G. G. & RYAN, J. W. 2022. Pharmaceutical innovation and access to financial markets. *PloS one*, 17.
- BERNSTEIN, S., LERNER, J. & MEZZANOTTI, F. 2019. Private Equity and Financial Fragility during the Crisis. *The Review of financial studies*, 32, 1309-1373.
- BOUCLY, Q., SRAER, D. & THESMAR, D. 2011. Growth LBOs. *Journal of financial economics*, 102, 432-453.
- BROCKER, M. 2024a. Brand Name Pharmaceutical Manufacturing in the US. IBIS
- BROCKER, M. 2024b. Generic Pharmaceutical Manufacturing in the US. IBIS
- BUSFIELD, J. 2020. Documenting the financialisation of the pharmaceutical industry. *Social Science & Medicine*, 258, 113096.

- COHEN, J. P. 2019 *How Recession Proof Is The Pharmaceutical Industry?* [Online]. Forbes Available: <https://www.forbes.com/sites/joshuacohen/2019/10/02/how-recession-proof-is-the-pharmaceutical-industry/> [Accessed 17 January 2025].
- DABBS, E. 2019. Why private equity is investing in biopharma. *MedNows*, 6-7.
- DAGGETT, P. 2024. *Why private equity is driving investment in healthcare* [Online]. FT Available: <https://www.ftadviser.com/investments/2024/10/04/why-private-equity-is-driving-investment-in-healthcare/> [Accessed January 17 2025].
- DE CONCINI, A., BRZEZICKA, PAULINA 2018. Access-to-finance conditions for Life Sciences R&D: What works and what needs fixing? : Innovation Finance Advisory, European Investment Bank Advisory Services.
- DIMASI, J. A., GRABOWSKI, H. G. & HANSEN, R. W. 2016. Innovation in the pharmaceutical industry: new estimates of R&D costs. *Journal of health economics*, 47, 20-33.
- FERNALD, K. D. S., FÖRSTER, P. C., CLAASSEN, E. & VAN DE BURG WAL, L. H. M. 2024. The pharmaceutical productivity gap – Incremental decline in R&D efficiency despite transient improvements. *Drug discovery today*, 29, 104160.
- FLOERSH, H. 2024. *Private equity invests in trial sites as CRO consolidation continues: PitchBook* [Online]. Available: <https://www.fiercebiotech.com/cro/private-equity-invests-trial-sites-cro-consolidation-continues-pitchbook> [Accessed January 20 2025].
- GOMPERS, P., KAPLAN, S. N. & MUKHARLYAMOV, V. 2016. What do private equity firms say they do? *Journal of Financial Economics*, 121, 449-476.
- GOOLD, F. B. A. M. 2007 The Strategic Secret of Private Equity. *Harvard Business Review*. Harvard Business Review
- HEMEL, D. J. & OUELLETTE, L. L. 2023. Valuing medical innovation. *Stanford law review*, 75, 517-599.
- HUR, W. & OH, J. 2021. A man is known by the company he keeps?: A structural relationship between backward citation and forward citation of patents. *Research policy*, 50, 104117.
- JOSE LUIS BLANCO, W. B., MATTHEW MALONEY & , A. J. P. 2024. *Bridging private equity's value creation gap* [Online]. Online McKinsey & Company Available: <https://www.mckinsey.com/industries/private-capital/our-insights/bridging-private-equitys-value-creation-gap#/> [Accessed 24 January 2025].
- KESSELHEIM, A. S. & AVORN, J. 2009. Using Patent Data to Assess the Value of Pharmaceutical Innovation. *The Journal of law, medicine & ethics*, 37, 176-183.
- LIGHT, D. & LEXCHIN, J. 2012. PHARMACEUTICAL R&D What do we get for all that money? *BMJ (Overseas and retired doctors ed.)*, 345, 22-25.

- M&A, O. 2025. ORBIS M&A: Deals: Pharmaceutical Manufacturers Moody's
- MAGRI, S. 2014. Does issuing equity help R&D activity? Evidence from unlisted Italian high-tech manufacturing firms. *Economics of innovation and new technology*, 23, 825-854.
- MELES, A., MONFERRÀ, S. & VERDOLIVA, V. 2014. Do the effects of private equity investments on firm performance persist over time? *Applied financial economics*, 24, 203-218.
- MOREIRA, S., KLUETER, T. M. & ASIJA, A. 2023. Market for technology 2.0? Reassessing the role of complementary assets on licensing decisions. *Research Policy*, 52, 104787.
- NEVELING, N. 2023. *Private Equity Locks in on Life Sciences* [Online]. Private Equity International Available: <https://www.privateequityinternational.com/private-equity-locks-in-on-life-sciences/> [Accessed January 2025].
- O'BRIEN, I. 2025. *Is pharma recession-proof?* [Online]. American Medical Journal Available: <https://www.emjreviews.com/en-us/amj/emj-gold/article/is-pharma-recession-proof/> [Accessed 22 January 2025].
- UPM, U. 2025. *What is a CDMO & How Can CDMO Services Streamline Your Pharma Development?* [Online]. Available: <https://www.upm-inc.com/what-is-a-cdm> [Accessed 22 January 2025].
- RAND 2021. The Financial Ecosystem of Pharmaceutical R&D. *Strategies in Regulated Markets (SiRM)*. RAND Corporation.
- SCHEFFLER RICHARD M AND ALEXANDER, L. M. A. G. J. R. 2021. Soaring Private Equity Investment in the Healthcare Sector: Consolidation Accelerated, Competition Undermined, and Patients at Risk.
- PPD Scientific, T. 2025. *What Is a CRO?* [Online]. Available: <https://www.ppd.com/what-is-a-cro/> [Accessed January 18 2025].
- SINGHAL, S., PATEL, N. & JAIN, A. 2025. *What to expect in US healthcare in 2025 and beyond* [Online]. McKinsey and Company Available: <https://www.mckinsey.com/industries/healthcare/our-insights/what-to-expect-in-us-healthcare-in-2025-and-beyond> [Accessed 28 January 2025].
- STATISTA, R. D. 2024. *Number of private equity (PE) deals in the United States from 2006 to 2023* [Online]. Statista. Available: <https://www.statista.com/statistics/377474/number-of-private-equity-deals-usa/> [Accessed 22 January 2025].
- SU, C.-Y., GUO, Y.-N., CHAI, K.-C. & KONG, W.-W. 2021. R&D Investments, Debt Capital, and Ownership Concentration: A Three-Way Interaction and Lag Effects on Firm Performance in China's Pharmaceutical Industry. *Frontiers in public health*, 9, 708832-708832.

- SUN, D., GAO, W., HU, H. & ZHOU, S. 2022. Why 90% of clinical drug development fails and how to improve it? *Acta pharmaceutica Sinica. B*, 12, 3049-3062.
- SVIRYDZENKA, K. 2016. Introducing a New Broad-based Index of Financial Development. *IMF Working Papers*, 16, 1.
- TYKVOVÁ, T. & BORELL, M. 2012. Do private equity owners increase risk of financial distress and bankruptcy? *Journal of corporate finance (Amsterdam, Netherlands)*, 18, 138-150.
- WALTON, J. Investors' views on Merger and acquisition, alliance and licensing activity in the pharmaceutical industry'. Consolidation and Competition in the Pharmaceutical Industry, based on papers delivered at the OHE Conference, London, 2000.
- WITTE, P. 2024. *Private Equity Pulse: key takeaways from Q3 2024* [Online]. EY Available: https://www.ey.com/en_us/insights/private-equity/pulse [Accessed 28 January 2025].
- YOUNKIN, P. 2008. Making the market: how the American pharmaceutical industry transformed itself during the 1940s. UC Berkeley: Center for Culture, Organizations and Politics

8. Appendix

Appendix A: Dependent Variable Calculation and Data Source

Variable Name	Variable in Data	Calculation	Data Source
Innovation Output	innovout	$\ln(1 + \text{patents quant})$	(Seru, 2023)
Value of Innovation Output	valinn	$\ln(i=1 \sum N(\text{Patent} \times \text{Forward citations}))$	(Seru, 2023)
R&D Intensity	lnrdint	$\ln(1 + (\text{R\&D expense} / \text{total assets}))$	Compustat
Altman Z score	lnzscore	$\ln(1 + 1.2 \times (\text{working capital} / \text{total assets}) + 1.4 \times (\text{retained earnings} / \text{total assets}) + 3.3 \times (\text{EBIT} / \text{total assets}) + 0.6 \times (\text{Common Equity} / \text{Liabilities total}) + 1 \times (\text{sale} / \text{total assets}))$	Compustat

Appendix B: Control Variables Calculations and Data Source

Variable	Acronym	Calculation	Data Source	Academic Source
Labor Productivity	lnLP	$\ln(\text{sale}/\text{employees})$	Compustat	(Ames et al., 2016)
Z Score	lnzscore	$\ln(1+1.2 \times (\text{working capital}/\text{total assets}) + 1.4 \times (\text{retained earnings}/\text{total assets}) + 3.3 \times (\text{EBIT}/\text{total assets}) + 0.6 \times (\text{Common Equity}/\text{Liabilities total}) + 1 \times (\text{sale}/\text{total assets}))$	Compustat	(Tykvová and Borell, 2012)
Return on Equity	roe	$\text{net income}/\text{total asset-total liabilities}$	Compustat	(Yang et al., 2024)
GDP Growth	gdp_growth	$(\text{GDP Year 2}-\text{Gdp Year 1})/\text{GDP year 1}$	FRED	(Tykvová and Borell, 2012).
Leverage Ratio	LR	$\text{liabilities total}/\text{total assets}$	Compustat	(Haque, 2023)
Financial Market Development	lnmarcap	$\ln(\text{Stock price} \times \text{total outstanding shares})$	CRSP	(Kelly, 2012)
Age	lnage	$\ln(2023-\text{IPOdate})$	Refinitiv	(Coad et al., 2018)
Tobin Q	Intobin	$\ln(\text{stock price} \times \text{csho} + \text{liabilities total})/\text{total assets}$	Compustat/CRSP	(Meles et al., 2014)
Employee Count	lnemp	$\ln(\text{employee count})$	Compustat	(Ames et al., 2016)
Institutional Ownership	instow	$\text{instow}/\text{total equity}$	Compustat	(Armitage et al., 2025).
Size	lnsize	$\ln(1+\text{total assets})$	Compustat	(Ames et al., 2016)
Capital Ratio	CR	$(\text{Stockholder Equity}/\text{Total Assets})$	Compustat	(Tykvová and Borell, 2012)
Cash Flow	CF	$((\text{operating activities net cash flow}/\text{total assets}))$	Compustat	(Kelly, 2012)
R&D Intensity	lnrdint	$\ln(\text{R\&D expense}/\text{total assets})$	Compustat	(Katila, 200)
Capital Productivity	lnCP	$\ln(\text{sale}/\text{Total PPE})$	Compustat	(Meles et al., 2014)

Appendix C: Pearson Correlation Matrix

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)
(1) Innovation Value	1.000																
(2) Innovation Output	0.562*	1.000															
(3) Labor Productivity	0.025	0.187*	1.000														
(4) Z-score	0.117*	0.165*	0.322*	1.000													
(5) Return on Equity	0.012	0.064*	0.221*	0.210*	1.000												
(6) GDP Growth	-0.077*	-0.013	0.007	0.072*	-0.017	1.000											
(7) Leverage Ratio	-0.097*	-0.067*	-0.004	-0.544*	0.310*	-0.072*	1.000										
(8) Financial Market Development	0.186*	0.378*	0.396*	0.534*	0.277*	-0.029	-0.090*	1.000									
(9) Age	0.246*	0.267*	0.075*	0.071*	0.086*	-0.140*	0.031*	0.091*	1.000								
(10) Tobin-Q	0.171*	0.161*	-0.063*	-0.117*	0.029	-0.093*	0.178*	0.360*	0.141*	1.000							
(11) Employee Count	0.254*	0.509*	0.402*	0.486*	0.187*	0.004	-0.124*	0.657*	0.198*	0.001	1.000						
(12) Institutional Ownership	0.145*	0.364*	0.352*	0.288*	0.126*	0.002	-0.077*	0.603*	-0.006	0.136*	0.506*	1.000					
(13) Size	0.246*	0.533*	0.430*	0.549*	0.114*	0.062*	-0.317*	0.740*	0.026	-0.062*	0.851*	0.686*	1.000				
(14) Capital Ratio	0.098*	0.067*	0.007	0.541*	0.310*	0.072*	-0.995*	0.089*	-0.027	-0.170*	0.122*	0.081*	0.314*	1.000			
(15) Cash Flow	0.128*	0.247*	0.544*	0.650*	0.273*	-0.019	-0.248*	1.000	0.281*	-0.064*	0.638*	0.290*	0.572*	0.245*	1.000		
(16) R&D Intensity	-0.005	0.034*	-0.332*	-0.570*	0.223*	-0.019	0.216*	0.091*	-0.195*	0.225*	-0.418*	-0.101*	-0.371*	-0.211*	-0.666*	1.000	
(17) Capital Productivity	-0.017	-0.001	0.716*	0.121*	0.138*	-0.050*	0.084*	0.360*	0.154*	0.029	0.062*	0.076*	0.020	-0.083*	0.315*	-0.227*	1.000

Significance: *** indicates p-value<0.01

Appendix D: Stata Code for T-Tests

```
/* T-Tests*/
ttest innovout, by(peba)
ttest valinn, by(peba)
ttest lnrdint, by(peba)
ttest lnzscore, by(peba)
```

Appendix E: Stata Code Model 1

```
*Innovaotion Output*
/*VIF*/
reg innovout peba lnLP lnzscore roe gdp_growth LR lnmarcap lnage lntobin lnemp instow lnsize CR CF
lnrdint lnCP reg_period fin_cycle tech_epoch
VIF
/* Random Effect model Choice*/
xttest0
/*Auto Correlation*/
xtserial innovout peba lnLP lnzscore roe gdp_growth LR lnmarcap lnage lntobin lnemp instow lnsize CF
lnrdint lnCP reg_period fin_cycle tech_epoch
/*Testing for Normality of Random Effects Shapiro-Wilk W Test on Predicted Random Effects*/
xtreg innovout peba lnLP lnzscore roe gdp_growth LR lnmarcap lnage lntobin lnemp instow lnsize CF lnrdint
lnCP i.reg_period i.fin_cycle i.tech_epoch, re vce(cluster firm_id)
predict u, u
swilk u
/*Final Reg Innovout*/
xtreg innovout peba lnLP lnzscore roe gdp_growth LR lnmarcap lnage lntobin lnemp instow lnsize CF lnrdint
lnCP i.reg_period i.fin_cycle i.tech_epoch, re vce(cluster firm_id)
```

Appendix F: Stata Regression Model 1

Random-effects GLS regression
Group variable: **firm_id**

Number of obs = **3,022**
Number of groups = **349**

R-squared:
Within = **0.0582**
Between = **0.3860**
Overall = **0.4493**

Obs per group:
min = **1**
avg = **8.7**
max = **38**

corr(u_i, X) = **0** (assumed)

Wald chi2(22) = **295.19**
Prob > chi2 = **0.0000**

(Std. err. adjusted for **349** clusters in **firm_id**)

innovout	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
peba	.2839752	.076403	3.72	0.000	.134228	.4337224
lnLP	.0008144	.0245278	0.03	0.974	-.0472592	.0488879
lnzscore	-.038221	.022444	-1.70	0.089	-.0822105	.0057685
roe	-.0088512	.0150667	-0.59	0.557	-.0383814	.0206789
gdp_growth	.00057	.0032977	0.17	0.863	-.0058933	.0070333
LR	.0718903	.0774986	0.93	0.354	-.0800042	.2237848
lnmarcap	-.0172257	.0355586	-0.48	0.628	-.0869192	.0524679
lnage	.7683102	.1154432	6.66	0.000	.5420456	.9945747
lntobin	.0601089	.0415166	1.45	0.148	-.021262	.1414799
lnemp	.0694222	.0360769	1.92	0.054	-.0012872	.1401315
instow	.0965322	.1211006	0.80	0.425	-.1408207	.3338851
lnsize	.1648402	.0486904	3.39	0.001	.0694086	.2602717
CF	-.076168	.0836115	-0.91	0.362	-.2400435	.0877075
lnrdint	.0963262	.0242652	3.97	0.000	.0487673	.1438852
lnCP	.0047078	.0201136	0.23	0.815	-.0347141	.0441297

Appendix G: Stata Code Model 2

```

*Innovaotion Value*
/*VIF*/
reg valinn peba lnLP lnzscore roe gdp_growth LR lnmarcap lnage lntobin lnemp instow lnsize CR CF lnrdint
lnCP reg_period fin_cycle tech_epoch
VIF
/* Random Effect model Choice*/
xtreg valinn peba lnLP lnzscore roe gdp_growth LR lnmarcap lnage lntobin instow lnsize CF lnrdint lnCP i.
reg_period i.fin_cycle i.tech_epoch, vce(cluster firm_id)
xttest0
/*Testing for Normality of Random Effects Shapiro-Wilk W Test on Predicted Random Effects*/
reg valinn peba lnLP lnzscore roe gdp_growth LR lnmarcap lnage lntobin instow lnsize CF lnrdint lnCP i.
reg_period i.fin_cycle i.tech_epoch, vce(cluster firm_id)
predict u, resid
swilk u
/*Auto Correlation*/
xtserial valinn peba lnLP lnzscore roe gdp_growth LR lnmarcap lnage lntobin instow lnsize CF lnrdint lnCP
reg_period fin_cycle tech_epoch
/*Final reg Valinn*/
reg valinn peba lnLP lnzscore roe gdp_growth LR lnmarcap lnage lntobin instow lnsize CF lnrdint lnCP i.
reg_period i.fin_cycle i.tech_epoch, vce(cluster firm_id)

```

Appendix H: Stata Regression Model 2

Linear regression	Number of obs	=	3,022
	F(21, 348)	=	26.07
	Prob > F	=	0.0000
	R-squared	=	0.2786
	Root MSE	=	.9597

(Std. err. adjusted for **349** clusters in **firm_id**)

valinn	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
peba	.2556728	.0707765	3.61	0.000	.1164694	.3948763
lnLP	-.0248066	.0372826	-0.67	0.506	-.0981342	.048521
lnzscore	-.0796623	.0283676	-2.81	0.005	-.1354558	-.0238689
roe	-.0137119	.0178971	-0.77	0.444	-.048912	.0214881
gdp_growth	-.0162766	.0049002	-3.32	0.001	-.0259143	-.0066389
LR	-.0364929	.0731795	-0.50	0.618	-.1804227	.1074369
lnmarcap	.0039335	.0347214	0.11	0.910	-.0643566	.0722237
lnage	.2937669	.1431406	2.05	0.041	.0122374	.5752965
lntobin	.0877936	.0590282	1.49	0.138	-.0283034	.2038906
instow	.1043639	.1184068	0.88	0.379	-.1285191	.3372469
lnsize	.1163449	.0316085	3.68	0.000	.0541772	.1785126
CF	.0059497	.0978954	0.06	0.952	-.1865914	.1984909
lnrdint	.1172715	.0300664	3.90	0.000	.0581368	.1764062
lnCP	.014493	.0314025	0.46	0.645	-.0472697	.0762557

Appendix I: Stata Code Model 3

```

*R&D Intensity*
/*VIF*/
reg lnrdint peba lnLP lnzscore roe gdp_growth LR lnmarcap lnage lntobin lnemp instow lnsize CR CF
lnrdint lnCP reg_period fin_cycle tech_epoch
VIF
/* Random Effect model Choice*/
xtreg lnrdint peba instow CF lnLP gdp_growth lnzscore lnmarcap lnCP CR lnage roe lnsize lntobin i.
tech_epoch i.fin_cycle i.reg_period, re
xttest0
/*Testing for Normality of Random Effects Shapiro-Wilk W Test on Predicted Random Effects*/
xtreg lnrdint peba instow CF lnLP gdp_growth lnzscore lnmarcap lnCP CR lnage roe lnsize lntobin i.
tech_epoch i.fin_cycle i.reg_period, re
predict u, resid
swilk u
/*Auto Correlation*/
xtserial lnrdint peba instow CF lnLP gdp_growth lnzscore lnmarcap lnCP CR lnage roe lnsize lntobin
reg_period fin_cycle tech_epoch
/*Final reg R&D Intensity*/
xtreg lnrdint peba instow CF lnLP gdp_growth lnzscore lnmarcap lnCP CR lnage roe lnsize lntobin i.
tech_epoch i.fin_cycle i.reg_period, re vce(cluster firm_id)

```

Appendix J: Stata Regression Model 3

Random-effects GLS regression	Number of obs	=	3,022
Group variable: firm_id	Number of groups	=	349
R-squared:	Obs per group:		
Within = 0.3736	min =		1
Between = 0.3902	avg =		8.7
Overall = 0.4150	max =		38
	Wald chi2(20)	=	503.61
corr(u_i, X) = 0 (assumed)	Prob > chi2	=	0.0000

(Std. err. adjusted for 349 clusters in firm_id)

lnrdint	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
peba	.5469997	.1006143	5.44	0.000	.3497992	.7442002
instow	.0186954	.0989851	0.19	0.850	-.175312	.2127027
CF	-.9011097	.0881887	-10.22	0.000	-1.073956	-.7282631
lnLP	.036471	.0357217	1.02	0.307	-.0335424	.1064843
gdp_growth	-.0001901	.003275	-0.06	0.954	-.006609	.0062289
lnzscore	-.0366715	.0190816	-1.92	0.055	-.0740707	.0007277
lnmarcap	-.1128854	.0391662	-2.88	0.004	-.1896497	-.036121
lnCP	-.0447209	.0319212	-1.40	0.161	-.1072853	.0178435
CR	-.0720105	.0811899	-0.89	0.375	-.2311398	.0871188
lnage	-.3878239	.1650381	-2.35	0.019	-.7112926	-.0643551
roe	-.0505908	.0138053	-3.66	0.000	-.0776487	-.0235329
lnsize	-.0741527	.0438516	-1.69	0.091	-.1601002	.0117947
lntobin	.2776607	.0585528	4.74	0.000	.1628994	.3924221

Appendix K: Stata Code Model 4

```

/*R&D Intensity with Debt*/
/*VIF*/
reg lnrdint peba pebalr lnLP lnzscore roe gdp_growth LR lnmarcap lnage lntobin lnemp instow lnsize CR CF
lnrdint lnCP reg_period fin_cycle tech_epoch
VIF
/* Random Effect model Choice*/
xtreg lnrdint peba pebalr instow CF lnLP gdp_growth lnzscore lnmarcap lnCP CR lnage roe lnsize lntobin i.
tech_epoch i.fin_cycle i.reg_period, re
xttest0
/*Testing for Normality of Random Effects Shapiro-Wilk W Test on Predicted Random Effects*/
xtreg lnrdint peba pebalr instow CF lnLP gdp_growth lnzscore lnmarcap lnCP CR lnage roe lnsize lntobin i.
tech_epoch i.fin_cycle i.reg_period, re
predict u, resid
swilk u
/*Auto Correlation*/
xtserial lnrdint peba pebalr instow CF lnLP gdp_growth lnzscore lnmarcap lnCP CR lnage roe lnsize lntobin
reg_period fin_cycle tech_epoch
/*Final reg R&D Intensity with Debt*/
xtreg lnrdint peba pebalr instow CF lnLP gdp_growth lnzscore lnmarcap lnCP CR lnage roe lnsize lntobin i.
tech_epoch i.fin_cycle i.reg_period, re vce(cluster firm_id)

```

Appendix L: Stata Regression Model 4

Random-effects GLS regression	Number of obs	=	3,022
Group variable: firm_id	Number of groups	=	349
R-squared:	Obs per group:		
Within = 0.3742	min =		1
Between = 0.3900	avg =		8.7
Overall = 0.4149	max =		38
	Wald chi2(21)	=	517.57
corr(u_i, X) = 0 (assumed)	Prob > chi2	=	0.0000

(Std. err. adjusted for **349** clusters in **firm_id**)

lnrdint	Robust					
	Coefficient	std. err.	z	P> z	[95% conf. interval]	
peba	.5982485	.1183918	5.05	0.000	.3662049	.8302922
pebalr	-.1132132	.1487509	-0.76	0.447	-.4047597	.1783332
instow	.0199311	.0986369	0.20	0.840	-.1733938	.2132559
CF	-.903271	.0873965	-10.34	0.000	-1.074565	-.7319769
lnLP	.0371993	.0356573	1.04	0.297	-.0326877	.1070863
gdp_growth	-.0001152	.0032522	-0.04	0.972	-.0064893	.006259
lnzscore	-.0377594	.0189827	-1.99	0.047	-.0749647	-.0005541
lnmarcap	-.113377	.0391625	-2.90	0.004	-.1901342	-.0366199
lnCP	-.0443933	.0317423	-1.40	0.162	-.106607	.0178205
CR	-.1410376	.1134301	-1.24	0.214	-.3633564	.0812812
lnage	-.3860146	.1649681	-2.34	0.019	-.7093462	-.0626829
roe	-.0499093	.0137627	-3.63	0.000	-.0768838	-.0229349
lnsize	-.0742385	.0438932	-1.69	0.091	-.1602676	.0117905
lntobin	.2775961	.0584333	4.75	0.000	.163069	.3921232

Appendix M: Stata Code Model 5

```

/*Z-Score*/
/*VIF*/
reg lnzscore peba pebalr lnLP lnzscore roe gdp_growth LR lnmarcap lnage lntobin lnemp instow lnsize CR
CF lnrdint lnCP reg_period fin_cycle tech_epoch
VIF
/* Random Effect model Choice*/
xtreg lnzscore peba instow CF lnrdint lnLP gdp_growth lnmarcap lnCP lnage roe LR lnsize lntobin i.
reg_period i.fin_cycle i.tech_epoch,re
xttest0
/*Testing for Normality of Random Effects Shapiro-Wilk W Test on Predicted Random Effects*/
xtreg lnzscore peba instow CF lnrdint lnLP gdp_growth lnmarcap lnCP lnage roe LR lnsize lntobin i.
reg_period i.fin_cycle i.tech_epoch,re
predict u, resid
swilk u
/*Auto Correlation*/
xtserial lnzscore peba instow CF lnrdint lnLP gdp_growth lnmarcap lnCP lnage roe LR lnsize lntobin
reg_period fin_cycle tech_epoch
/*Final reg Z-Score with Debt*/
xtreg lnzscore peba instow CF lnrdint lnLP gdp_growth lnmarcap lnCP lnage roe LR lnsize lntobin i.
reg_period i.fin_cycle i.tech_epoch,re vce(cluster firm_id)

```

Appendix N: Stata Regression Model 5

Random-effects GLS regression	Number of obs	=	3,022
Group variable: firm_id	Number of groups	=	349
R-squared:			
Within	=	0.1185	
Between	=	0.2748	
Overall	=	0.2756	
	Obs per group:		
	min	=	1
	avg	=	8.7
	max	=	38
	Wald chi2(20)	=	205.34
corr(u_i, X) = 0 (assumed)	Prob > chi2	=	0.0000

(Std. err. adjusted for **349** clusters in **firm_id**)

lnzscore	Robust					[95% conf. interval]
	Coefficient	std. err.	z	P> z		
peba	-.0635546	.061991	-1.03	0.305	-.1850547	.0579455
instow	-.2017251	.1045774	-1.93	0.054	-.406693	.0032429
CF	.0626035	.0784659	0.80	0.425	-.0911868	.2163938
lnrdint	-.0862161	.029541	-2.92	0.004	-.1441154	-.0283167
lnLP	-.0611157	.0279628	-2.19	0.029	-.1159218	-.0063097
gdp_growth	-.0048984	.0048586	-1.01	0.313	-.014421	.0046242
lnmarcap	.0892017	.0304857	2.93	0.003	.0294508	.1489525
lnCP	.0664736	.0234586	2.83	0.005	.0204955	.1124516
lnage	-.0392392	.1148152	-0.34	0.733	-.2642728	.1857944
roe	.063195	.0150415	4.20	0.000	.0337141	.0926759
LR	-.6744036	.1013924	-6.65	0.000	-.8731292	-.4756781
lnsize	.0634741	.0294949	2.15	0.031	.005665	.1212831
lntobin	.0143378	.0440463	0.33	0.745	-.0719913	.1006669

Appendix O: Wooldridge Test for Auto Correlation Robustness Check

Variable	Assumption	Test-Statistic	p-value	Result
Innovation Output	No first-order autocorrelation	$F(1, 260) = 16.349$	0.0001	Autocorrelated
Innovation Value	No first-order autocorrelation	$F(1, 260) = 13.776$	0.0003	Autocorrelated
R&D Intensity	No first-order autocorrelation	$F(1, 260) = 79.946$	0.0000	Autocorrelated
R&D Intensity with Interaction	No first-order autocorrelation	$F(1, 260) = 80.110$	0.0000	Autocorrelated
Z-Score	No first-order autocorrelation	$F(1, 260) = 46.380$	0.0000	Autocorrelated

Appendix P: Breusch-Pagan Lagrange multiplier for Random Effects Robustness Check

Variable Model	Assumption	X^2	p-value	Result
Innovation Output	The variance of the random effects is zero	4497.25	0.000	Random Effects is preferred over OLS
Innovation Value	The variance of the random effects is zero	0	1.000	OLS is preferred over random effects
R&D Intensity	The variance of the random effects is zero	5967.32	0.000	Random Effects is preferred over OLS
R&D Intensity with Interaction	The variance of the random effects is zero	5962.82	0.000	Random Effects is preferred over OLS
Z-Score	The variance of the random effects is zero	1720.92	0.000	Random Effects is preferred over OLS

Appendix Q: Shapiro-Wilk Test for Normality Robustness Check

Variable	Assumption	Shapiro-Wilk Statistic (W)	p-value	Result
Innovation Output	The data is normally distributed	0.96982	0.0000	Data is not normally distributed
Innovation Value	The data is normally distributed	0.94941	0.0000	Data is not normally distributed
R&D Intensity	The data is normally distributed	0.9231	0.0000	Data is not normally distributed
R&D Intensity with Interaction	The data is normally distributed	0.92297	0.0000	Data is not normally distributed
Z-Score	The data is normally distributed	0.93163	0.0000	Data is not normally distributed

Appendix R: Balance Statistics Before and After Matching

Variable	Sample	Treated Mean	Control Mean	%Bias	%Red Bias	t-test (t)	p-value
Labor Productivity	U	279.13	278.94	0.1	-	0.02	0.986
	M	279.13	314.72	-10.7	-17997	-3.47	0.001
Leverage Ratio	U	0.42263	0.44406	-5.4	-	-1.65	0.099
	M	0.42263	0.45506	-8.2	-51.3	-2.79	0.005
Institutional Ownership	U	0.56083	0.35285	65.2	-	19.87	0
	M	0.56083	0.56759	-2.1	96.7	-0.67	0.5
Size	U	854.5	1798.5	-33.3	-	-10.62	0
	M	854.5	827.15	1	97.1	0.46	0.648
Z-Score	U	-1.9981	-1.9759	-0.2	-	-0.07	0.946
	M	-1.9981	-2.4735	4.7	-2042.7	1.55	0.121
R&D Intensity	U	0.31175	0.27307	12	-	3.72	0
	M	0.31175	0.30542	2	83.6	0.65	0.513
Cash Flow	U	-0.27809	-0.267	-2.3	-	-0.73	0.468
	M	-0.27809	-0.26983	-1.7	25.6	-0.6	0.551
Capital Ratio	U	0.572	0.54808	5.9	-	1.8	0.071
	M	0.572	0.53954	8	-35.7	2.72	0.006
Age	U	2.8043	2.8367	-4.4	-	-1.35	0.177
	M	2.8043	2.8078	-0.5	89.2	-0.17	0.865
Market Capitalization	U	19.73	19.345	1.5	-	0.48	0.634
	M	19.73	20.547	-3.3	-111.9	-1.12	0.262

U: Unmatched, M: Matched

Appendix S: Summary Statistics for Matching Quality

Sample	Pseudo R ²	LR Chi ²	p-val	Mean Bias	Median Bias	B (Multivariate Bias)	R (Variance Ratio)	% Reduction in Bias
Unmatched	0.145	756.97	0	12.3	4.5	96.1*	1.1	82
Matched	0.005	27.71	0.004	3.9	2.1	15.9	1.07	55

Appendix T: Propensity Score Matching Stata Code

```

/*Propensity Score Matching Innovout*/
use rd11.dta, clear
global covariates LP gdp_growth LR instow size zscore rdint CF CR age markcap
probit peba $covariates i.year i.reg_period i.fin_cycle i.tech_epoch
xtset firm_id year
predict ps, pr
egen year_group = group(year)
gen pscore_composite = year_group * 10 + ps
psmatch2 peba, pscore(ps) caliper(0.1) neighbor(1)
pstest $covariates, both
keep if _weight != .
xtset firm_id year
tsfill, full

```

Appendix U: Propensity Score Matched Model 1

Variable	Coefficient	Estimated % Change in Innovation Output	Significance
PE Backed Status	0.3030	35.39%	***
Labor Productivity	0.0160	1.60%	
Z Score	-0.0390	-3.82%	
Return on Equity	-0.0050	-0.50%	
GDP Growth	0.0010	0.10%	
Leverage Ratio	0.0980	9.80%	
Financial Market Development	-0.0140	-1.39%	
Age	0.8770	140.37%	***
Tobin Q	0.0600	6.18%	
Employee Count	0.0670	6.93%	*
Institutional Ownership	0.0870	9.09%	
Size	0.1720	18.77%	***
Cash Flow	-0.1070	-10.15%	
R&D Intensity	0.1190	12.64%	***
Capital Productivity	-0.0050	-0.50%	

Significance: * indicates p-value<0.1, ** indicates p-value<0.05, *** indicates p-value<0.0

Appendix V: Propensity Score Matched Model 2

Variable	Coefficient	Estimated % Change in Value of Innovation	Significance
PE Backing	0.2170	24.23%	***
Labor Productivity	0.0110	1.11%	
Z-Score	-0.0890	-8.5%	***
Return on Equity	-0.0110	-1.10%	
GDP Growth	-0.0180	-1.80%	***
Leverage Ratio	-0.0320	-3.20%	
Financial Market Development	0.0110	1.11%	
Age	0.2460	27.89%	***
Tobin-Q	0.0890	9.31%	
Institutional Ownership	0.0680	6.80%	
Size	0.1210	12.86%	***
Cash Flow	-0.0270	-2.70%	
R&D Intensity	0.1290	13.77%	***
Capital Productivity	-0.0080	-0.80%	

Significance: * indicates p-value<0.1, ** indicates p-value<0.05, *** indicates p-value<0.01

Appendix W: Propensity Score Matched Model 3

Variable	Coefficient	Estimated % Change in R&D Intensity	Significance
PE Backed Status	0.4020	49.48%	***
Institutional Ownership	0.0490	4.90%	
Cash Flow	-0.8610	-86.10%	***
Labor Productivity	0.0370	3.77%	
GDP Growth	0.0003	0.03%	
Z-Score	-0.0490	-4.78%	**
Financial Market Development	-0.1270	-11.93%	***
Capital Productivity	-0.0460	-4.50%	
Capital Ratio	-0.0380	-3.80%	
Age	-0.4710	-37.56%	***
Return on Equity	-0.0490	-4.90%	***
Size	-0.0590	-5.73%	
Tobin-Q	0.2840	32.84%	***

Significance: * indicates p-value<0.1, ** indicates p-value<0.05, *** indicates p-value<0.01

Appendix X: Propensity Score Matched Model 4

Variable	Coefficient	Estimated % Change in R&D intensity	Significance
PE Backed Status	0.4250	53.0%	***
PE-Backed Interaction	-0.0500	-4.9%	
Institutional Ownership	0.0500	5.00%	
Cash Flow	-0.8620	-86.20%	***
Labor Productivity	0.0370	3.77%	
GDP Growth	0.0003	0.03%	
Z-Score	-0.0500	-4.9%	**
Financial Market Development	-0.1280	-12.01%	***
Capital Productivity	-0.0460	-4.50%	
Capital Ratio	-0.0740	-7.40%	
Age	-0.4710	-37.56%	***
Return on Equity	-0.0490	-4.90%	***
Size	-0.0590	-5.73%	
Tobin-Q	0.2850	32.98%	***

Significance: * indicates p-value<0.1, ** indicates p-value<0.05, *** indicates p-value<0.01

Appendix Y: Propensity Score Matched Model 5

Coefficient	Significance	Estimated % Change in Z-Score	Significance
PE Backed Status	-0.074	-7.1%	
Institutional Ownership	0.0110	1.1%	*
Cash Flow	-0.0890	-8.9%	
R&D Intensity	-0.0110	-1.1%	***
Labor Productivity	-0.0180	-1.8%	**
GDP Growth	-0.0320	-3.2%	
Financial Market Development	0.0110	1.1%	***
Capital Productivity	0.2460	27.9%	***
Age	0.0890	9.3%	
Return on Equity	0.0680	6.8%	***
Leverage Ratio	0.1210	12.1%	***
Size	-0.0270	-2.7%	
Tobin-Q	0.1290	13.8%	

Significance: * indicates p-value<0.1, ** indicates p-value<0.05, *** indicates p-value<0.01

